Chapter 9
MOS Transistors and Field Controlled Wide Bandgap Devices

9.1 Function Principle of the MOSFET

The MOSFET basic structure was investigated early [Hof63]. For the comprehension of the function of a MOSFET (Metal Oxide Semiconductor Field Effect Transistor), the surface of the semiconductor may be examined at first. The surface of a semiconductor is always a disturbance of the ideal lattice due to the lack of neighboring atoms. Therefore, a thin oxide will always be built up on the surface or other atoms and molecules are adsorbed. Thus, these surface layers are normally electrically charged.

A p-type semiconductor may be given as an example. Assumed is a positive charge on the surface (Fig. 9.1).

For a small positive charge we obtain

$$|qV_S| < E_i - E_F:$$

The hole-concentration/density on the surface is being reduced. The conduction band and the valence band are bent downwards. A depletion zone of the thickness h_d is built up.

A higher positive charge yields

$$|qV_S| > E_i - E_F:$$

In this case, the conduction and valence band are bent even stronger. In a small region at the surface the Fermi level is now closer to the conduction band than to the valence band. An inversion layer of the thickness h_i is being formed, in which electrons are the majority charge carriers.

Next to this is the depletion layer with the thickness h_d, which separates the inversion layer from the p-type area.

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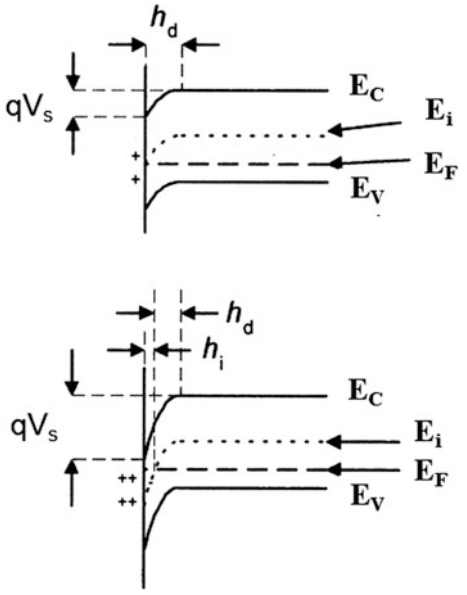


Fig. 9.1 Semiconductor surface. p-type semiconductor, positive charge on the surface

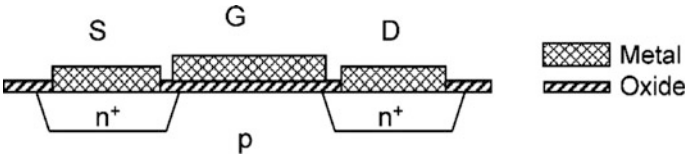


Fig. 9.2 Lateral n-channel MOSFET. Description: *S* Source, *G* Gate, *D* Drain

Having a negative charge on the p-type semiconductor, an accumulation layer of holes will be formed. The n-type semiconductor behaves similarly: With a positive charge at the surface, the accumulation layer is formed, with a negative surface charge a depletion zone is formed, with an increased negative charge the inversion layer.

Next, we assume a thin oxide film on the p-type semiconductor and apply a metallization on it. On this metal film a positive voltage is applied. Furthermore, two n^+-areas are added and bonded as source and drain region. We have the simplest case of a lateral MOS field effect transistor shown in Fig. 9.2 [Hof63].

A positive voltage has the same effect as the positive surface charge: When a sufficient positive voltage at the gate is applied, both n-areas are connected by the inversion layer. Due to the gate voltage $V_G > V_T$ a current can flow between the drain and the source.

Gate-Source Threshold-Voltage V_T (n-channel-MOSFET):
The threshold voltage *is the gate voltage, at which the generated electron concentration equals the concentration of the acceptors*.

It has to be distinguished:

n-channel MOSFET: An n-type channel is formed in a p-area.
p-channel MOSFET: A p-type channel is formed in a n-area.

On closer inspection, it has to be considered that the oxide contains positive charges on the boundary surface to the semiconductor. These charges are in the order of 5×10^9 to 1×10^{11} cm^{-2}. Moreover, the gate area of power-MOSFETs consists of a heavily n-doped poly silicon layer (see Figs. 9.4 and 9.5) and a potential difference already exists between gate and semiconductor due to the differing positions of the Fermi level in the n^+-doped poly silicon and in the p-type semiconductor (concerning the n-channel MOSFET). Both effects function in the same way as an external positive gate-voltage and result in a reduction of the threshold voltage V_T. In case of a low doping of the p-area and a high oxide charge of the n-channel MOSFET, V_T is negative, even without gate voltage a channel exists. The definition of the threshold voltage mentioned above remains valid.

It has to be distinguished:

Depletion type: $V_T < 0$. The device is normally on and does not block before a negative gate-source-voltage $V_G < V_T$ is applied.
Enhancement type: $V_T > 0$. An n-channel only develops when $V_G > V_T$ (normally off device).

Usually, MOSFETs of the enhancement type are used in power electronics, because of the normally-off feature that is required in voltage source converters. Almost always n-channel MOSFETs are used, being more advantageous since the mobility of the electrons is much higher than that of the holes (see Chap. 2). Typically, the threshold voltage of modern devices is adjusted between 2 and 4 V.

9.2 Structure of Power MOSFETs

The configuration shown in Fig. 9.2 will sustain little drain-source voltage. Thus, a structure such as in Fig. 9.3, which is named DMOS (D = double diffused), is used from 10 V upwards. The lowly doped n^--area, the drain extension area, takes over the blocking voltage.

Lateral DMOS transistors are frequently used in power ICs and in monolithic integrated power semiconductor-circuits ("smart power"). But they have the disadvantage of having a low current-load capacity, because the n^--area consumes a large part of the surface of the semiconductor. If real "power" has to be controlled, a

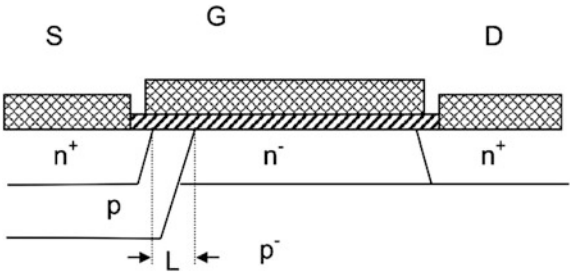


Fig. 9.3 Lateral DMOS

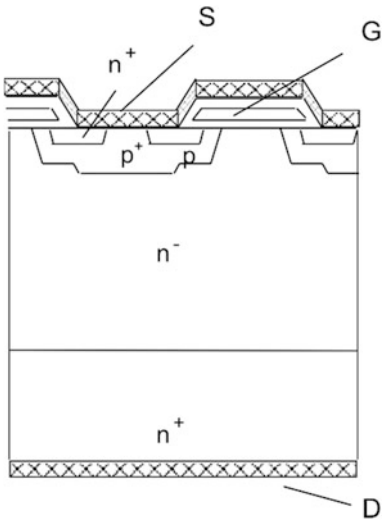


Fig. 9.4 Vertical DMOS transistor. The gate electrode is poly-silicon

vertical MOSFET is realized by arranging the area for the space charge vertically [Lid79], see Fig. 9.4. Consequently, the volume of the semiconductor is utilized and the surface can be used for the formation of the gate-source cell structure.

On the surface of the semiconductor the individual cells are formed, which consist of p-wells and n^+-source areas. A cross section of a cell can be seen in Fig. 9.4. The p-well is connected to the source metallization, so that the parasitic npn-transistor is shorted. In order that the short has very little resistance, the doping is increased at this spot by an additional p^+-implantation, followed by a diffusion step. At the edges of the well is the channel, which is covered with the thin gate oxide. Above the oxide the gate electrode is applied, usually consisting of a heavily doped n^+ poly-silicon. At one spot, mostly in the center of the chip, the gate electrode is brought to the surface to form a gate pad to which a bond wire can be attached.

Because the current has to flow through the inversion channel, many single cells are formed to gain a larger width of the channel. An example is given in Fig. 9.5. Here, the cells are of square form and arranged in a square pattern. The

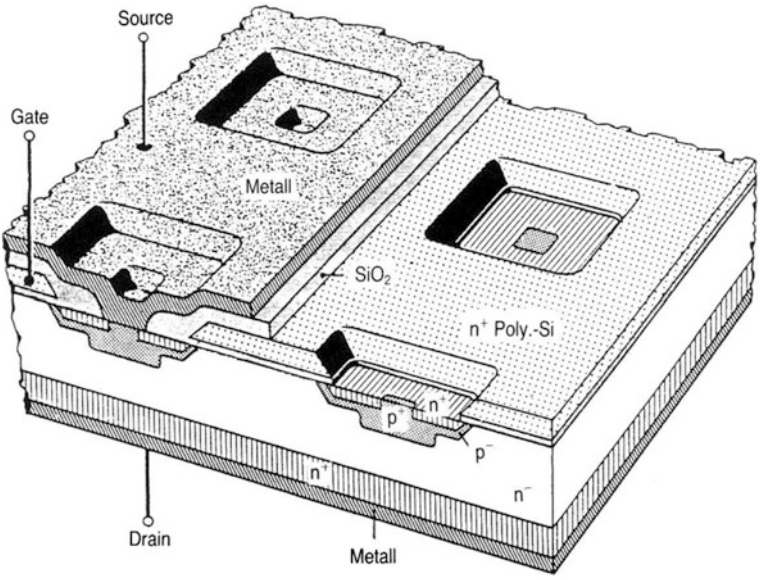


Fig. 9.5 Cell structure of a vertical DMOS. From [Ste92]

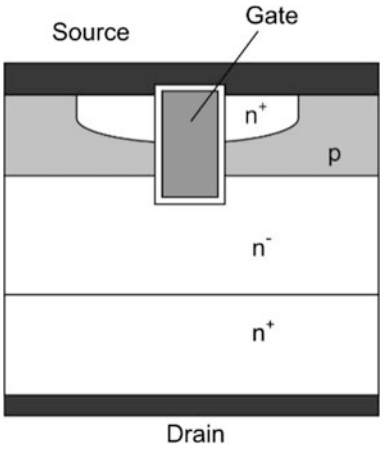


Fig. 9.6 Vertical trench-MOSFET

semiconductor surface area is used even more effectively with a hexagonal pattern, where the single cells are hexagonal, the so-called HEXFET structure [Col79].

The vertical DMOS (also called VDMOS) transistors are used in a wide field of applications, such as computer and electronic equipment power supplies with power factor correction front ends. Since the second half of the 1990s, with the introduction of the trench-MOS [Sod99] a further improvement has been made, in which the channel area is vertically arranged (Fig. 9.6). Due to this, a much smaller on-state resistance can be gained especially in the lower voltage range < 100 V.

9.3 Current-Voltage Characteristic of MOS-Transistors

Figure 9.7 shows the current-voltage characteristics of the MOSFET. The device is in the blocking state as long as a positive voltage V_D between drain and source is given and V_G is smaller than the threshold voltage V_T. The blocking voltage of the MOSFET is limited by the avalanche breakdown. Because the npn-transistor is suspended by a low resistance short, the blocking voltage of the MOSFET corresponds to the blocking voltage of the diode, which is formed by the p-well, the low doped base region and the n^+-layer.

A current carrying channel is built for $V_G > V_T$, resulting in the given current-voltage characteristics. Similar to the current gain of bipolar transistors a transconductance is defined here. For low voltages V_D, the current-voltage characteristics have the form of a straight line. For a defined gate voltage V_G the resistance $R_{DS(on)}$ (= R_{on}) is indicated.

The transition between the ohmic region and the pinch-off region is called the quasi saturation. This region is described by a parabolic curve.

In the reverse direction of the MOSFET a forward biased diode is present. As for a power diode, this diode forward characteristic is often approximated by a threshold voltage V_{F0} and a differential resistance.

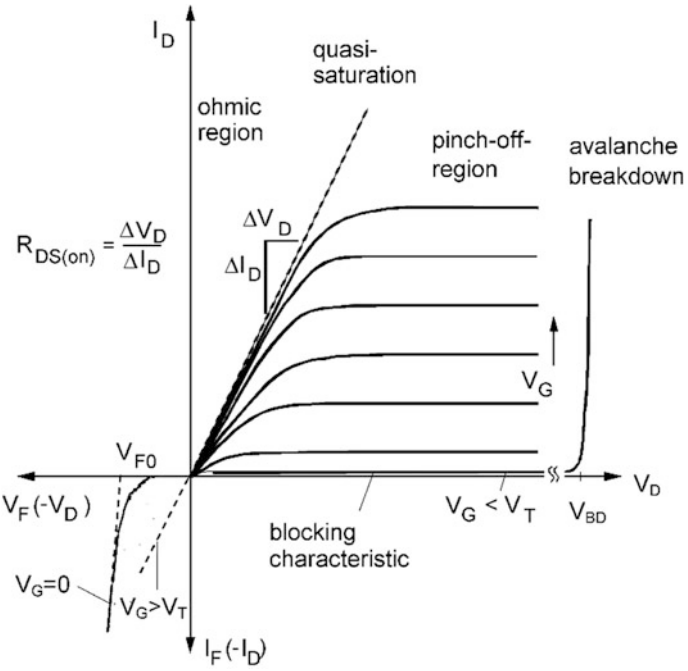


Fig. 9.7 Current-voltage characteristic of the MOSFET. Figure adapted from [Nic00]

9.4 Characteristics of the MOSFET Channel

Due to the oxide layer of the gate and the gate electrode a capacitor is built above the channel. It's area specific capacity can be described with

$$C_{ox} = \frac{\varepsilon_0 \cdot \varepsilon_r}{d_{ox}} \tag{9.1}$$

The oxide has the thickness d_{ox} (i.e. < 100 nm). The relative permittivity of the oxide is ε_r = 3.9 (for SiO_2). Having the gate voltage V_G higher than the threshold voltage V_T an inversion channel is created, as it is shown in Fig. 9.8a. As long as the voltage drop caused by the current in the channel can be neglected, the charge of the inversion channel yields

$$Q_s = C_{ox} \cdot (V_G - V_T) \tag{9.2}$$

The carriers, forming this charge, are available for the current transport in the inversion channel. As long as the pinch-off can be neglected in the channel, the resistance of the channel is

$$R_{ch} = \frac{L}{W \cdot \mu_n \cdot Q_s} = \frac{L}{W \cdot \mu_n \cdot C_{ox} \cdot (V_G - V_T)} = \frac{1}{\kappa \cdot (V_G - V_T)} \tag{9.3}$$

where L is the length of the channel (for example 2 µm, see Fig. 9.3) and W is the entire width of it. In Fig. 9.3, W is vertical to the plane of projection and corresponds to the circumference of the single cell multiplied by the number of cells (see Fig. 9.5). Having a high cell density, a large W is achieved, and thus a low channel resistance. W can amount to some 100 m per cm^2 of a chip surface in modern semiconductor devices. The parameters dependent on geometry can be summarized in

$$\kappa = \frac{W \cdot \mu_n \cdot C_{ox}}{L} \tag{9.4}$$

Equation (9.3) is valid for the ohmic region in Fig. 9.7; that is for the area in which the voltage drop across the channel can be neglected with regard to its influence on Q_S.

As can be seen in (9.3), the channel resistance is affected by the carrier mobility. In Sect. 2.6 it has been shown, that the mobility μ_p only amounts to approximately one third of μ_n. It is for this reason that, whenever it is possible, n-channel MOSFETs are used in power electronics.

With increasing current a voltage drop $V(y)$ develops across the channel. The channel narrows, see Fig. 9.8b. Along the length of the channel y a charge $Q(y)$ will exist. For an element dR of the resistance R_{CH}, with (9.3) one obtains

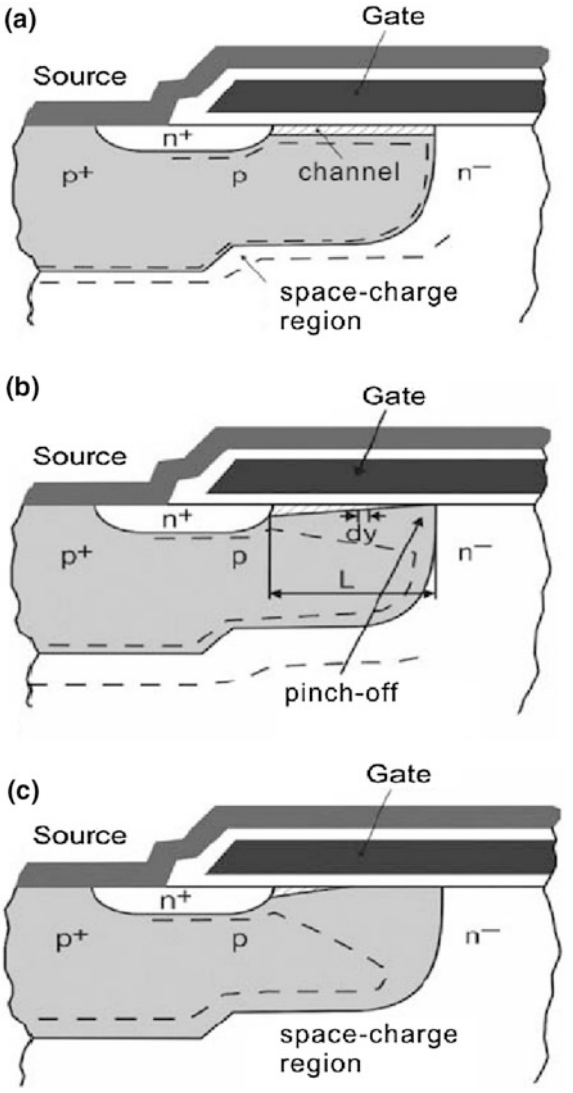


Fig. 9.8 MOSFET channel. **a** Ohmic region, $V_D \ll V_G - V_T$, **b** pinch-off, $V_D = V_G - V_T$, **c** channel length shortening $V_D \gg V_G - V_T$

$$dR = \frac{dy}{W \cdot \mu_n \cdot Q(y)} \tag{9.5}$$

with

$$Q(y) = C_{ox} \cdot (V_G - V_T - V(y)) \tag{9.6}$$

In a segment dR the voltage drop is

$$dV = I_D \cdot dR \tag{9.7}$$

Inserting (9.6) and (9.5) in (9.7) yields

$$I_D = W \cdot \mu_n \cdot C_{ox} \cdot (V_G - V_T - V(y)) \cdot \frac{dV}{dy} \tag{9.8}$$

The voltage V_D drops between the boundaries $y = 0$ and $y = L$:

$$\int_0^L I_D \cdot dy = W \cdot \mu_n \cdot C_{ox} \cdot \int_0^{V_D} (V_G - V_T - V(y)) \cdot dV \tag{9.9}$$

Integration leads to the following $I_D(V_G,V_D)$ characteristic:

$$I_D = \kappa \cdot \left((V_G - V_T) \cdot V_D - \frac{1}{2} V_D^2 \right) \tag{9.10}$$

for $V_D \leq V_G - V_T$. The characteristic corresponds to the parabolic section (quasi saturation) in Fig. 9.7. For small V_D it verges into

$$I_D = \kappa \cdot (V_G - V_T) \cdot V_D \tag{9.11}$$

and corresponds to the ohmic region, as already indicated in (9.3). The passing into the pinch-off region results from Eq. (9.10) for $dI_D/dV_D = 0$. Afterwards the channel is pinched off for

$$V_D = V_G - V_T \tag{9.12}$$

For a larger V_D, inserting (9.12) in (9.10) yields the characteristics in the pinch-off region. In this region, the current remains almost constant even for increased voltage V_D

$$I_{Dsat} = \frac{\kappa}{2} \cdot (V_G - V_T)^2 \tag{9.13}$$

The transconductance is defined by

$$g_{fs} = \left. \frac{\Delta I_D}{\Delta V_G} \right|_{V_D = const} \tag{9.14}$$

By differentiating (9.13) we obtain

$$g_{fs} = \kappa \cdot (V_G - V_T) \tag{9.15}$$

According to (9.13), the current I_{Dsat} is independent of V_D. However, in reality the electric field penetrates into the p-zone, when V_D is strongly increased (see Fig. 9.8c) and the channel becomes shortened. This shortening of the channel-length involves a slight ascent of the current at high voltages.

The current-voltage characteristics (9.10) can be found in numerous textbooks. However, comparing it with practically realized power devices, this equation is not very satisfactory. In the derivation it was not considered that a depletion zone is formed below the channel. This zone widens while the channel is narrowing, as it is indicated in Fig. 9.8c. A derivation of the current-voltage characteristics in consideration of the space charge can be found in [Gra89, Sze81]. This leads to

$$I_D = \kappa \cdot \left((V_G - V_T) \cdot V_D - \frac{1}{2} \left(1 + \frac{C_D}{C_{ox}} \right) V_D^2 \right) \tag{9.16}$$

for $V_D \leq V_G - V_T$. It contains the area specific capacity of the space charge region

$$C_D = \sqrt{\frac{\varepsilon_0 \cdot \varepsilon_r \cdot q \cdot N_A}{2 \cdot \Delta V_T}} \tag{9.17}$$

as it has been derived for the treatment of the pn-junction with the Eq. (3.109). The voltage ΔV_T in Eq. (9.17) corresponds to the voltage, which is necessary to develop a space charge region in the p-doped well with doping concentration N_A:

$$\Delta V_T = 2 \cdot \frac{k \cdot T}{q} \cdot \ln \left(\frac{N_A}{n_i} \right) \tag{9.18}$$

ΔV_T is necessary to fulfill the condition of strong inversion. ΔV_T amounts to approximately 0.81 V (resulting from a typical doping of the p-well of 1×10^{17} cm^{-3}). Considering the space charge this way, little changes at a very small voltages V_D. The approximation for the ohmic region in Eq. (9.11) remains the same. However, I_{Dsat} and g_{fs} vary. Equation (9.16) is effective as long as the voltage drop across the channel is smaller than ΔV_T that is $V_D < \Delta V_T$.

Apart from that, the reduced mobility in the channel has to be considered. Even without a lateral electric field, the mobility is already reduced compared to the values indicated in Fig. 2.12 of Chap. 2. The reason is the influence of the semiconductor surface. If a voltage $V(y)$ is formed above the channel, then a significant electric field develops in lateral direction. Equation (2.38) has to be consulted for the velocity of the electrons. For the electron mobility it yields

$$\mu_{nCH} = \frac{\mu_{e0}}{1 + \theta \cdot (V_G - V_T)} \tag{9.19}$$

In [Gra89] a suitable agreement with experiments is achieved, when the values of 600 $cm^2V^{-1}s^{-1}$ and 0.02 V^{-1} are used for μ_{e0} and θ.

The threshold voltage for the MOSFET is expressed with [Sze81]

$$V_T = V_{p,S} - \frac{Q_{ox}}{C_{ox}} + \Delta V_T + \frac{\sqrt{2qN_A\varepsilon_0\varepsilon_r\Delta V_T}}{C_{ox}} \tag{9.20}$$

with the potential difference n^+-poly/p-Si

$$V_{p,S} = \frac{kT}{q}\ln\frac{N_{D,poly} \cdot N_A}{n_i^2} \tag{9.21}$$

and the charge due to oxide traps Q_{ox}. The threshold voltage is temperature dependent and decreases with temperature primarily due to the strong temperature dependency of n_i in (9.18) and (9.21).

Measurements of the threshold voltage are in most cases executed with shorted gate and drain and applying a small current I_D. The found temperature dependency with this method is stronger than predicted by Eq. (9.20). Main reason is that at low currents the condition of strong inversion is not fulfilled. If for sufficient high drain voltages ($V_D > 5$ V) V_T is extracted at the intercept of the extrapolated tangent of $\sqrt{I_D}$ versus V_G at $V_{GS} \gg V_T$ [Lee82], a reasonable agreement with (9.20) is found.

9.5 The Ohmic Region

For the MOSFET's ohmic resistance not only the channel resistance has to be considered. Indeed, the resistance of the low doped middle region already dominates in devices with a blocking voltage of 50 V upwards. Because this layer is grown by epitaxy for vertical MOSFETs, the designation R_{epi} is commonly used. Figure 9.9 describes the structure of the MOSFET with a given path of the charge carriers (electrons) and with different parts of the resistance

$$R_{on} = R_{S^*} + R_{n+} + R_{ch} + R_a + R_{epi} + R_s \tag{9.22}$$

For MOSFETs with blocking voltages < 50 V, effort is made to reduce the channel resistance. The near-surface parts are reduced by increased cell density (larger W, see Eq. (9.1)). Most progress is achieved with the trench-cell (Fig. 9.5), where additionally the resistance R_a is eliminated.

In Table 9.1, the shares of the particular parts of the resistance are specified for a 30 V vertical MOSFET with planar cells together with a corresponding 600 V MOSFET.

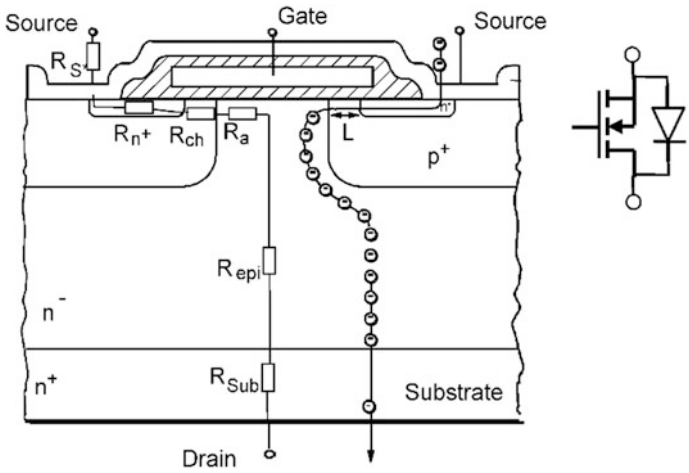


Fig. 9.9 Current path and resistances in a MOSFET. Figure inspired by [Lor99]

Table 9.1 Resistance R_{on} for MOSFETs with different blocking voltages

		V_{DS} = 30 V	V_{DS} = 600 V
R_S*	Package	7%	0.5%
R_{n+}	Source layer	6%	0.5%
R_{CH}	Channel	28%	1.5%
R_a	Accumulation layer	23%	0.5%
R_{epi}	n^--layer	29%	96.5%
R_{Sub}	Substrate	7%	0.5%

Values from [Lor99]

The resistance R_{epi} of the low doped region is identical with the voltage drop across the low doped base region of a unipolar device, which has been given in Chap. 6 for Schottky diodes (Eq. 6.8):

$$R_{epi} = \frac{w_B}{q \cdot \mu_n \cdot N_D \cdot A} \tag{9.23}$$

If the device is designed for higher voltages, w_B has to be chosen larger as well as N_D smaller. Having a conventional MOSFET, the resistance can be calculated according to the shown approach in dependence of the voltage for which the device is designed. The lowest resistance can be obtained for a light PT design, as indicated in Eq. (6.14):

$$R_{epi,\min} = 0.88 \cdot \frac{2 \cdot B^{\frac{1}{2}} \cdot V_{BD}^{\frac{5}{2}}}{\mu_n \cdot \varepsilon \cdot A} \tag{9.24}$$

Thus, the resistance increases more than with the square of the blocking voltage, namely with $V_{BD}^{2.5}$. Equations (9.23) and (9.24) or comparable equations (see Chap. 6) are handled as "unipolar limit" in the literature. Meanwhile, this limit has been broken by the principle of compensation structures.

9.6 Compensation Structures in Modern MOSFETs

The compensation principle for power MOSFETs has been introduced in commercially available products in 1998 with the 600-V CoolMOS™ technology [Deb98]. The basic principle behind the drastic $R_{on} \cdot A$ reduction compared to conventional power MOSFETs is the compensation of n-drift region donors by acceptors located in p-columns (also known as superjunction). Figure 9.10 shows the structure of a superjunction MOSFET compared to a conventional MOSFET. In the middle layer, p-columns are arranged. Their p-doping is adjusted to the value necessary for compensation of the n-regions. The compensating acceptors are located in lateral proximity to the drift region donors.

The result is a low effective doping in the entire voltage-sustaining region. An almost rectangular shape of the electric field is obtained, as can be seen in the lower part of Fig. 9.10. For this field shape, the highest voltage can be absorbed at a given

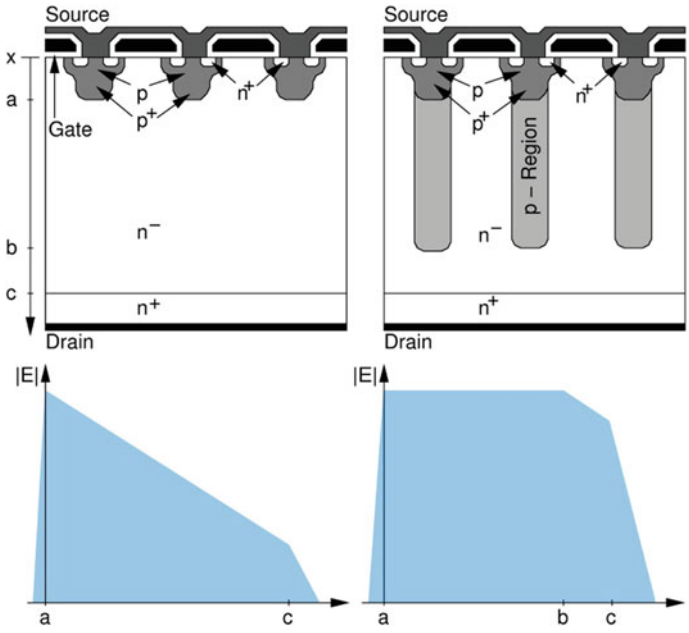


Fig. 9.10 Standard MOSFET and superjunction MOSFET

thickness. The doping of the n-layer can be lifted in so far, as it is technologically possible to compensate it through an equally large p-doping. In this process it has to be considered that the area of the n- respectively n^--region is decreased.

By means of the compensation principle, the coupling of the blocking voltage and the doping is neutralized and a degree of freedom for the adjustment of the n-doping is obtained. Since, according to (9.23), the n-doping determines the resistance in unipolar devices, the resistance can be lowered drastically.

In case of the rectangular shape of the electric field, the avalanche breakdown can be calculated by using the ionization integral (3.71), with the approach proposed by Shields and Fulop with $n = 7$ (see the sections about the PT diode, Eq. (5.12)), resulting in

$$w_B = B^{\frac{1}{6}} \cdot V_{BD}^{\frac{7}{6}} \tag{9.25}$$

with $B = 2.1 \times 10^{-35} \mathrm{cm}^6 \mathrm{V}^{-7}$ as derived in Eq. (3.83). Inserting (9.25) into (9.23) yields

$$R_{epi} = \frac{2 \cdot B^{\frac{1}{6}} \cdot V_{BD}^{\frac{7}{6}}}{q \cdot \mu_n \cdot N_D \cdot A} \tag{9.26}$$

The factor 2 in the numerator of (9.26) is obtained due to the simplified account that the width of the p-columns is equal to the width of the n-zones. Only the n-areas contribute to the conduction, hence only half of the area is available.

Figure 9.11 compares the relation between R_{epi} and the blocking voltage for the conventional design (9.24) and for the superjunction device (9.26). Here, the doping $N_A = N_D = 2 \times 10^{15}\,\mathrm{cm}^{-3}$ is chosen for the superjunction device. Furthermore, half of the total area is assumed to carry the electron current.

The following consequences can be derived for this very simplified case.

Under blocking conditions the space charge laterally penetrates into the n- and p-region. This is shown in Fig. 9.12. In Fig. 9.12 it is assumed that p- and n-region

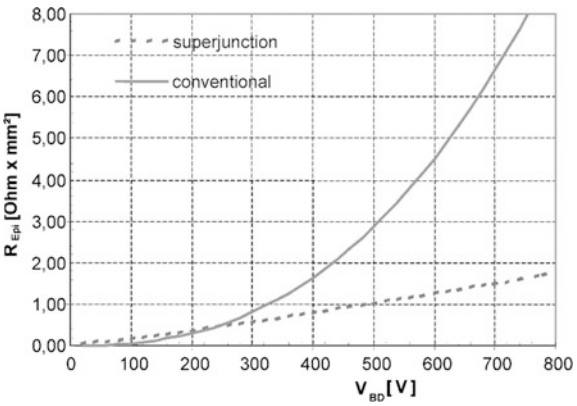


Fig. 9.11 Resistance R_{epi} as a function of the blocking voltage for the conventional MOSFET and for the superjunction device

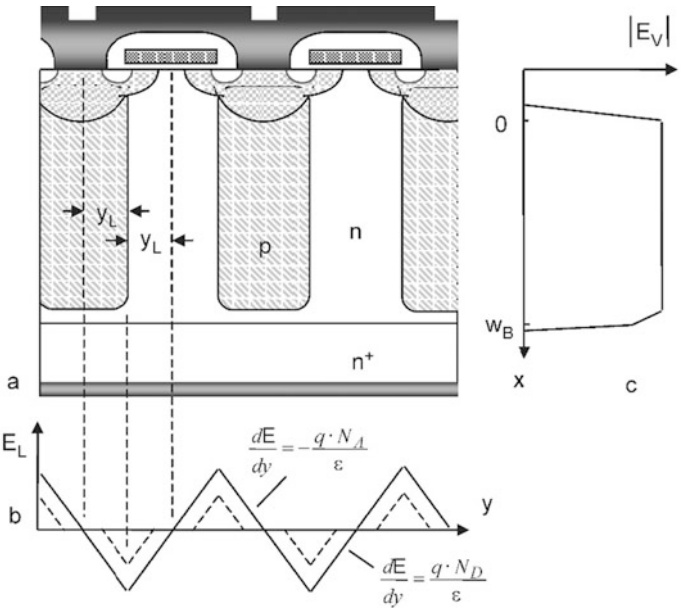


Fig. 9.12 Superjunction MOSFET. **a** Simplified structure **b** electric field in lateral direction in the region of the columns **c** electric field in vertical direction

are doped equally; the p-region has the doping N_A, the n-column the doping N_D with $N_A = N_D$. Moreover, both columns shall feature the same width, which constitutes $2 \cdot y_L$ respectively in Fig. 9.12. When a voltage is applied in reverse direction, at first the space charge region penetrates only laterally into the columns. For low voltages, the dashed line in Fig. 9.12b indicates the magnitude of the electric field along a section in lateral direction in the region of the columns. With increasing voltage, the space charges will eventually meet in the center of the respective columns, as shown by the solid line in Fig. 9.12b. Now all acceptors and donors are ionized.

With further increase of the voltage the zigzag line in Fig. 9.12b is lifted. This results in a structure similar to a corrugated iron roof. In vertical direction the electric field as shown in Fig. 9.12c is obtained.

In lateral direction, the respective p- and n-regions have to be penetrated entirely by the field. The expansion of the electric field at the avalanche breakdown into an n-region with the doping N_D has been given in Eq. (3.85). The indicated width w equals half the n-region y_L. A doping $N_D = 2 \times 10^{15}\,\mathrm{cm}^{-3}$ yields $y_L = 11\ \mu\mathrm{m}$. The width of the p- as well as the n-regions has to be smaller than $2 \cdot y_L$, otherwise breakdown occurs in lateral direction.

Therefore, the doping in Eq. (9.26) is connected to the width of the columns; a higher doping N_D demands a smaller y_L. Equation (3.85) inverted for N_D and inserted in Eq. (9.26) yields

$$R_{epi} = \frac{2 \cdot 2^{-\frac{3}{7}} B^{\frac{13}{42}} \cdot y_L^{\frac{8}{7}} \cdot V_{BD}^{\frac{7}{6}}}{\varepsilon \cdot \mu_n \cdot A} \tag{9.27}$$

Analogous considerations are leading to a similar result in [Zin01]. Equation (9.27) shows that the resistance can be reduced even more as shown in Fig. 9.11. This requires a yet smaller y_L. Note that the line for the superjunction in Fig. 9.11 can be shifted to a lower value for R_{epi} for an increased doping and a finer pattern. However, to realize this in the vertical structure with a depth $w_B \gg y_L$ is a major technological challenge.

A more precise consideration, also including peaks of the electric field at the source and drain sided border of the space charge region, can be found in [Che01]. There, different arrangements of the columns are analyzed as well. Seen from the top of a device, the considerations of Figs. 9.10 and 9.12 would result in a stripe pattern of the p- and n-regions. However, a hexagonal arrangement could also be suitable.

The requirement for precise lateral n- and p-dose compensation limits the n-drift region doping. The higher the deviation in the charge balance, the more loss in blocking capability occurs. This effect increases with higher doping N_D and smaller y_L. The process window for deviation from charge balance gets narrower [Kon06]. Process technology finally limits the possibility to reduce R_{on} in this type of compensation power MOSFETs.

For breakdown voltages below 200 V, field-plate or oxide-bypassed MOSFETs are an excellent alternative [Lia01, Sie06c]. The device comprises a deep trench penetrating most of the n-drift region. An isolated field plate provides mobile charges required to compensate the drift region donors under blocking conditions as shown in Fig. 9.13. A voltage source dynamically provides electrons on the field plate and therefore precise lateral drift region compensation is ensured under all operating conditions.

The field plate isolation has to withstand the full source-drain blocking voltage of the device at the trench bottom; therefore oxide layers with thickness in the micron range have to be fabricated carefully with a special focus on avoiding thinning at the bottom trench corners and preventing generation of stress-induced

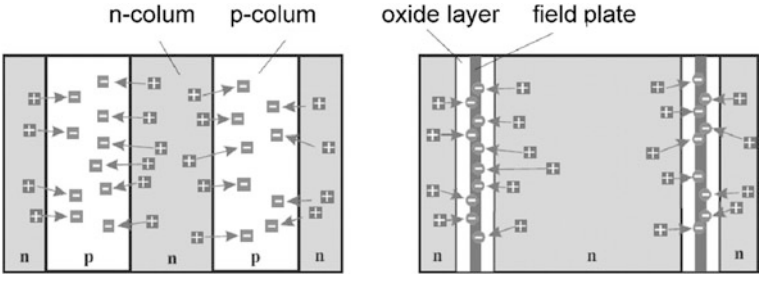


Fig. 9.13 Comparison of charge compensation by superjunction and by field-plates

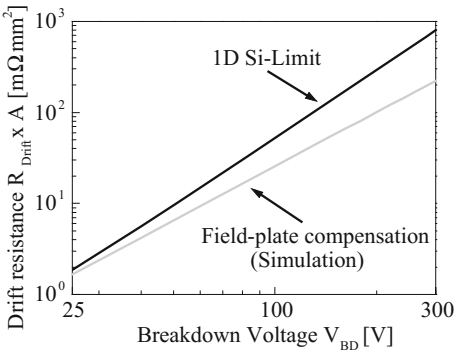


Fig. 9.14 Drift region resistance in dependency of breakdown voltage for field-plate devices, compared to the "silicon limit"

defects. In contrast to standard trench MOS structures that exhibit a linearly decreasing electric field with a maximum at the body/drift region pn-junction, the field-plate principle leads to a more constant field distribution. While in superjunction devices an almost homogeneous vertical field distribution is realized, the shape of the electric field of a field-plate device shows two peaks, one at the body/drift region pn-junction and the larger one at the bottom of the field-plate trench [Che05]. The necessary drift region length for a given breakdown voltage is reduced and the drift region doping can be increased, leading to a significantly reduced on-state resistance.

As in case of superjunction devices, the on-state resistance is reduced below the "Silicon Limit". Depending on the feasible device geometry, either superjunction or field-plate devices are advantageous in terms of device performance. In the voltage range between 30 and 100 V, the field plate compensation is superior to the superjunction compensation [Che05]. Figure 9.14 depicts the drift region resistance R_{epi} as a function of blocking voltage. A comparison is given between 2D Simulations [Paw08] for field plate compensation structures and the "silicon limit" given by Eq. (9.24). Due to the small device dimensions of the field-plate trench devices, the consequence is a larger doping density and thus a smaller drift region resistance [Paw08]. The simulation results in Fig. 9.14 agree well with experimental data.

9.7 Temperature Dependency of MOSFET Characteristics

The threshold voltage V_T given in (9.20) decreases with temperature. On the other hand, the mobility μ_n decreases with temperature, and it is included in all main factors that contribute to $R_{(on)}$ in Eq. (9.22). R_{on} increases with temperature. Since, for example, μ_n (125 °C) is approx. $0.5 \cdot \mu_n$ (25 °C), R_{on} doubles.

In the transfer characteristics is shown in Fig. 9.15 [Wil17]. At low V_G the saturation current I_{Dsat} increases with T due to the dominating influence of decreasing V_T. At high V_G, I_{Dsat} decreases due to the dominating mobility decrease. The intersection point is denoted as temperature compensation point TCP.

The temperature coefficient $ß_T$ is given by

$$ß_T = \frac{I_D(T_2) - I_D(T_1)}{\Delta T} \tag{9.28}$$

at $V_G = const$. Usually power MOSFETs are not designed to be operated in the pinch-off region, also denoted as linear region, see Fig. 9.7. If done, then operation below the TCP may lead to thermal instability. For small gate voltages, the drain current increases with temperature due the temperature dependency of the threshold voltage. If the device is operated within this region, a hot spot will develop and a thermal runaway occurs as shown in Fig. 9.16. At larger gate voltages, the drain current decreases with increasing temperature because the carrier mobility reduces at higher temperature. The larger the channel width, the more pronounced is the region of instability.

Figure 9.16 gives an example and shows temperature transients of a power MOSFET device at different voltage bias while keeping the pulse power at a the same value [Spi02]. Whether the device is stable or destroyed depends entirely on the bias conditions, i.e., the drain voltage, the drain current, and the duration of the pulse, not simply on the average power. For example, as shown in Fig. 9.16, a 90-W pulse power application realized as 6 A with V_D = 15 V is stable but at V_D = 30 V and I_D = 3 A, an operation point below the TCP, the device goes into destructive thermal runaway. Using the temperature coefficient of current $ß_T = \Delta I_D/\Delta T$ the operation points can be distinguished. Depending on the bias, the device remains stable of a hot spot grows.

At higher cell densities and large channel width W, as given in modern trench MOSFETs, the safe operating region for linear circuit mode is reduced [Cha16], whereby even relatively short durations of current saturation during start-up or shutdown may lead to device failure [Wil17].

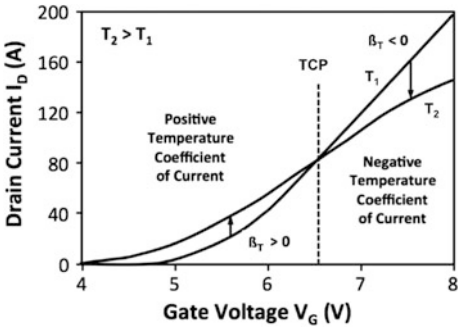


Fig. 9.15 Transfer characteristics at $V_D = const.$, $V_D > V_G - V_T$, at two different temperatures. Figure from R. Siemieniec, Infineon

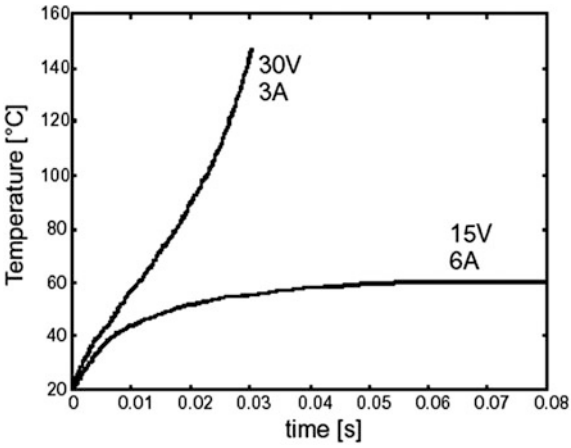


Fig. 9.16 Temperature transients of a power MOSFET at constant pulse power. Figure from P. Spirito, UniNa, redrawn from [Spi02]

9.8 Switching Properties of the MOSFET

Starting from the transition time of the charge carriers through the channel

$$\tau_t = \frac{L}{v_d} \tag{9.29}$$

with $v_d = \mu_n \cdot E$ and $E = V_{ch}/L$ yields

$$\tau_t = \frac{L^2}{\mu_n \cdot V_{ch}} \tag{9.30}$$

For instance, with $d = 2$ µm, $V_{ch} = 1$ V and $\mu_n = 500\,\mathrm{cm^2V^{-1}s^{-1}}$, we obtain the transition time $\tau_t \approx 80$ ps. It corresponds to a transition frequency of

$$f_t \approx 12.5\,\mathrm{GHz}$$

In practice, this is not accomplishable for a power MOSFET, since parasitic capacitances exist, leading to time-constants that determine the limiting frequency:

$$f_{co} = \frac{1}{2\pi \cdot C_{iss} \cdot R_G} \tag{9.31}$$

with $C_{iss} = C_{GS} + C_{GD}$ and $R_G = R_{G\mathrm{int}} + R_{Gext}$. C_{iss} as well as the recommended gate resistance R_{ext} can be found in the data sheets. The internal gate resistance has to be asked for from the manufacturer.

Example: IXYS XFH 67 N10

$$C_{iss} = 4500\,\text{pF}$$
$$R_{ext} = 2\Omega, R_{\text{int}} \approx 1\,\Omega\,(\text{assumed})$$
$$\Rightarrow f_{co} = 12\,\text{MHz}$$

Figure 9.17 shows the structure of the MOSFET in which the parasitic capacitances are indicated. On the right, the equivalent circuit diagram of the MOSFET with its parasitic capacitances is shown. The inverse diode and some of the resistances are illustrated as well, only R_{CH} and R_{epi} are charted.

The turn-on and turn-off behavior shall be dealt with now under the condition of an inductive load, as an inductive load is usually existent in practice. The circuit corresponds to the one in Fig. 5.19. Figure 9.18 shows the turn-on waveform of the MOSFET with an inductive load. The characteristic quantities for the turn-on are:

t_d: *Turn-on delay time*
time until V_G reaches the threshold voltage V_T

$$t_d \sim R_G \cdot (C_{GS} + C_{GD})$$

t_{ri}: *Rise time*

During this time the current increases

$$t_{ri} \sim R_G \cdot (C_{GS} + C_{GD})$$

Due to the freewheeling diode for the inductive load, the reverse current peak I_{RRM} is added, see Figs. 5.20 and 5.21. During this time the voltage V_D remains virtually unaltered.

t_{fv}: *Voltage fall time*

Now the freewheeling diode takes over voltage and the voltage across the MOSFET drops. The capacitance C_{GD} (Miller capacitance) is being charged.

$$t_{fv} \sim R_G \cdot C_{GD}$$

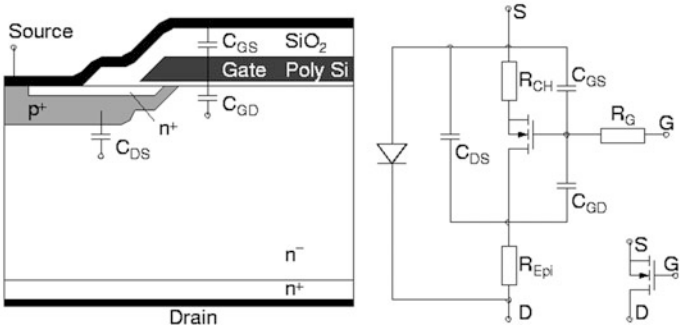


Fig. 9.17 MOSFET with parasitic capacitances, structure and electrical equivalent circuit according to [Mic03] © 2003 Springer

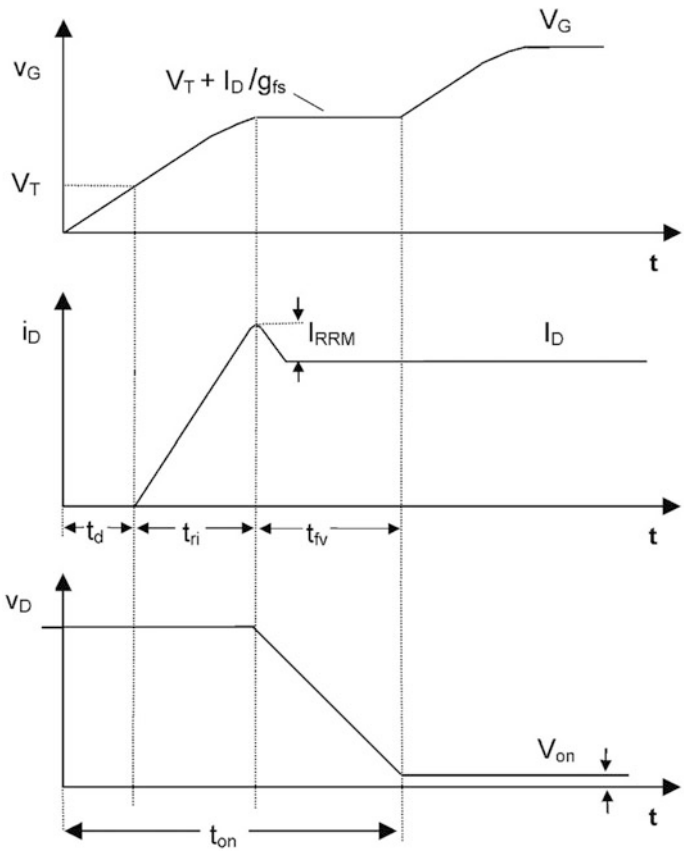


Fig. 9.18 Turn-on waveforms of the MOSFET with inductive load

In this phase, V_G remains at the value of the Miller plateau

$$V_G = V_T + I_D/g_{fs}$$

The voltage V_D falls to the value of the forward voltage

$$V_{on} = R_{on} \cdot I_D$$

The entire turn-on time t_{on} amounts to

$$t_{on} = t_d + t_{ri} + t_{fv}$$

Figure 9.19 shows the turn-off behavior for an inductive load. Characteristic quantities of the turn-off process are:

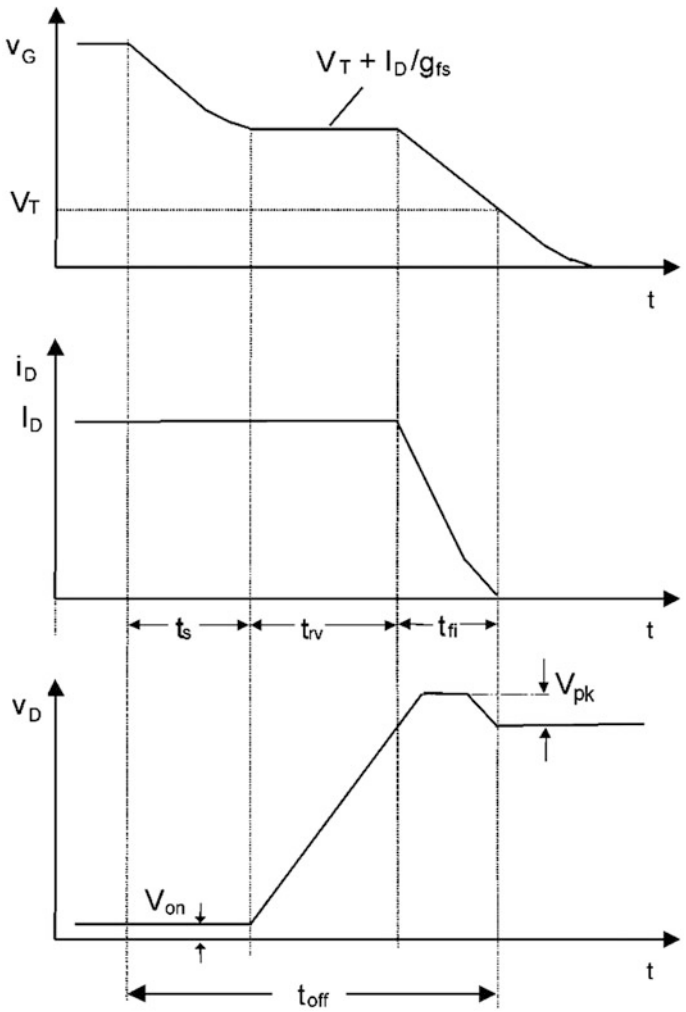


Fig. 9.19 Turn-off waveform of the MOSFET with inductive load

t_s: Storage time
In the driver the voltage signal is reset to zero or a negative value. However, the gate has to be discharged to the value at which the gate voltage corresponds to the value at which the on-state current I_D equals the saturation current, which means

$$V_G = V_T + I_D/g_{fs}$$

The capacitances C_{GS} and C_{GD}, which are in parallel to the channel, have to be discharged (see Fig. 9.15), For the storage time holds:

$$t_s \sim R_G \cdot (C_{GS} + C_{GD})$$

t_{rv}: *Voltage rise time*
The voltage increases to the value given by the circuit. The current remains constant at the initial value. The gate voltage persists at the Miller plateau. The Miller capacitance C_{GD} has to be discharged and, therefore

$$t_{rv} \sim R_G \cdot C_{GD}$$

t_{fi}: *Current fall time*
The gate capacitance C_{GS} + C_{GD} is discharged and the current decreases. The current becomes zero (or, more exact, it attains the value of the off-state leakage current), when V_{GS} has fallen to V_T.

$$t_{fi} \sim R_G \cdot (C_{GS} + C_{GD})$$

In this phase a spike V_{pk} is added to the applied voltage. This spike consists of

- the inductive voltage, which is generated by the current slope *di/dt* at the parasitic inductance L_{par}, L_{par} is indicated in Fig. 5.18.
- the turn-on dynamic forward voltage spike V_{FRM} of the diode

Consequently

$$V_{pk} = \left| L_{par} \cdot \frac{di}{dt} \right| + V_{FRM}$$

The entire turn-off time is

$$t_{off} = t_s + t_{rv} + t_{fi}$$

The switching edges of the turn-on and turn-off can be controlled by means of the gate resistance under the conditions described. With a smaller R_G the switching time can be reduced and, thus, the switching losses in the device may be lowered as well.

From the switching times a frequency limit can be derived

$$f_{\max} = \frac{1}{t_{on} + t_{off}} \tag{9.32}$$

Taking the data sheet values of the above mentioned MOSFET IXYS IXFH 67 N10 and adding all switching times typically 220 – 340 ns is obtained, which corresponds to a frequency of 3 – 4 MHz. This is considerably lower than f_{co}. But the example is not unproblematic, as the switching times in the data sheets are mostly specified for an ohmic load, which is rarely the case in practice.

9.9 Switching Losses of the MOSFET

The maximum attainable switching frequency of a power MOSFET depends on the switching losses. The energy loss per pulse can be calculated, like for other devices, by integrating the product $v(t) \cdot i(t)$ during turn-on and turn-off. During turn-on it can be calculated with

$$E_{on} = \int_{t_{on}} v_D(t) \cdot i_D(t) dt \tag{9.33}$$

In practice, the energy loss per pulse is determined from oscillograms. Modern oscilloscopes are able to calculate the product of current and voltage and to integrate it over the selected time. An example for an IGBT is given in Fig. 5.21. For an estimation, Fig. 9.18 can be used from which emanates

$$E_{on} = \frac{1}{2} \cdot V_D \cdot (I_D + I_{RRM}) \cdot t_{ri} + \frac{1}{2} \cdot V_D \left(I_D + \frac{2}{3} I_{RRM} \right) \cdot t_{\mathrm{fv}} \tag{9.34}$$

assuming that the peak reverse current I_{RRM} caused by the diode decays linear during the time t_{fv}.

At turn-off, the energy loss is calculated with

$$E_{off} = \int_{t_{off}} v_D(t) \cdot i_D(t) dt \tag{9.35}$$

which can be estimated according to Fig. 9.19 with

$$E_{off} = \frac{1}{2} \cdot V_D \cdot I_D \cdot t_{rv} + \frac{1}{2} \cdot \left(V_D + V_{pk} \right) \cdot I_D \cdot t_{fi} \tag{9.36}$$

The total switching losses follow from

$$P_{on} + P_{off} = f \cdot \left(E_{on} + E_{off}\right) \tag{9.37}$$

Conduction losses and blocking losses add to the switching losses. For power MOSFETs the off-state leakage current is in the order of few µA, so that the blocking losses may be neglected. The conduction losses cannot be ignored. Defining the duty-cycle d as the ratio of the interval in which the MOSFET conducts versus the switching period, the conduction losses can be calculated according to

$$P_{cond} = d \cdot V_{on} \cdot I_D = d \cdot R_{on} \cdot I_D^2 \tag{9.38}$$

For the total losses one obtains

$$P_V = P_{cond} + P_{on} + P_{off} = d \cdot R_{on} \cdot I_D^2 + f \cdot \left(E_{on} + E_{off}\right) \tag{9.39}$$

These losses have to be conducted away as heat flux through the case of the device. The maximum allowable losses are determined by the cooling conditions, the acceptable temperature difference and the thermal resistance. Details are provided in Chap. 11.

For the MOSFET IXYS IXFH 67 N10 used as an example, it can be estimated from the data sheet values of the thermal resistance that switching frequencies up to 300 kHz can be realized. Clearly, the MOSFET, as it is a unipolar device, is the fastest Si power semiconductor switch available.

The potential switching frequency depends, on the one hand, on the thermal parameters and considerably on the other devices in the circuit as well. The entire circuit has to be optimized accordingly. From (9.34) and (9.36) arises that the switching losses depend on the switching times. By a reduction of the switching times due to smaller gate resistances R_G, switching losses can be lowered. On the other hand, the steepness of the slopes is limited in practice

- by motor windings which are not to be stressed with too high *dv*/*dt*
- even more by freewheeling diodes, which are required in inductive circuits. Inappropriate freewheeling diodes, in the presence of increased *di*/*dt*, lead to a snappy switching behavior, voltage spikes and oscillations.

9.10 Safe Operating Area of the MOSFET

Between source and drain, the structure of the MOSFET contains a parasitic bipolar npn-transistor in parallel to the MOS channel, as shown in Fig. 9.20. This parasitic npn-transistor could lead to many problems:

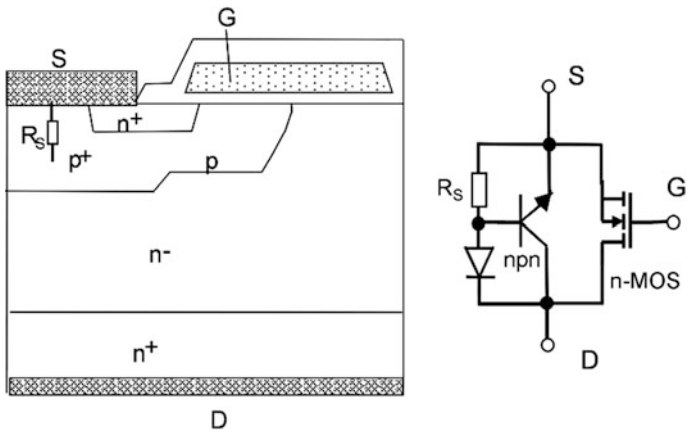


Fig. 9.20 MOSFET with its equivalent circuit, containing the parasitic npn-transistor and the parasitic diode

- the blocking voltage would be decreased by such an open base transistor
- when applying a voltage with a high *dv*/*dt*, a displacement current could be generated due to the charging of the depletion layer of the base-collector junction. This current triggers the transistor,
- and finally, the safe operating area (SOA) of a transistor is limited by the second breakdown effect.

Therefore, the base-emitter junction of the npn-transistor has to be shorted by a low resistance R_S. This resistance is chosen preferably small by increasing the doping in this region with an additional p^+ ion implantation (p^+ doping) and by choosing the length of the n^+ source region as small as possible, as the photolithographic process allows it.

In today's MOSFETs the parasitic transistor is effectively made inoperative. Thus, the safe operating area is no longer limited by the second breakdown. The safe operating area of today's MOSFETs is rectangular, as it is shown in Fig. 9.21. It is only limited by the blocking voltage and the occurring losses. The SOA curves for pulse times above 10 µs in Fig. 9.21 are limited due to the maximum power losses, at which the junction temperature remains below 150 °C.

9.11 The Inverse Diode of the MOSFET

Because of the contact between the p-well and the source metallization, a pin-type diode structure is built by the p-well, the n^- region and the n^+-substrate, as shown in Fig. 9.20. Consequently, for the application in a bridge topology in a voltage source converter circuit, a freewheeling diode is intrinsically present. The characteristics of

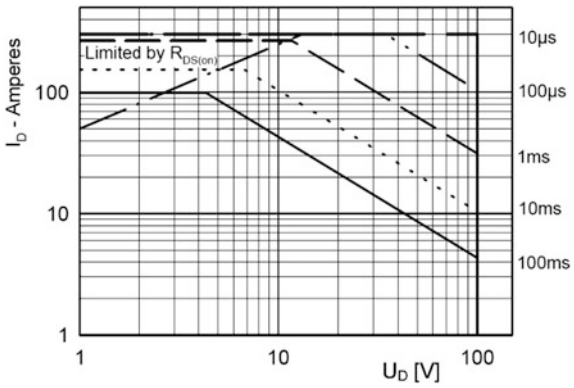


Fig. 9.21 Safe operating area (SOA) of a MOSFET, example IXYS IXFH 67 N10

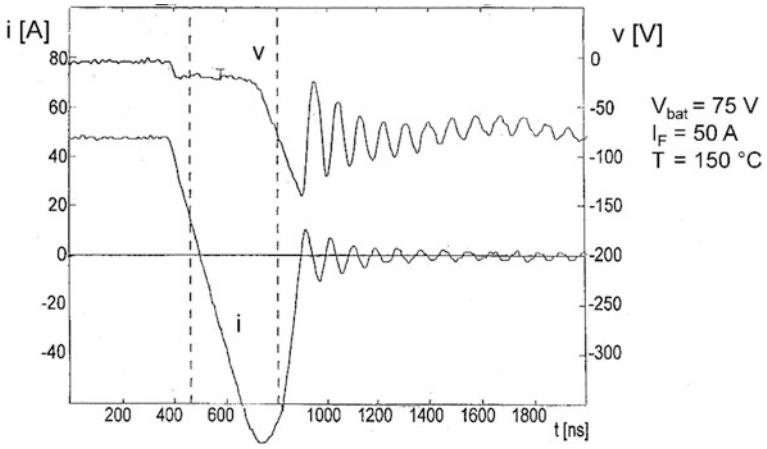


Fig. 9.22 Switching behavior of the inverse diode of a 200 V MOSFET with high-frequency LC-oscillations

this diode corresponds to the current-voltage characteristic of the MOSFET in the 3rd quadrant (see Fig. 9.7 for $V_G = 0$). However, the turn-off behavior of this intrinsic diode is relatively poor, compared to optimized pin-type or Schottky barrier diodes for the same blocking voltage. Figure 9.22 shows a snappy turn-off event of the inverse diode in a 200 V-MOSFET.

The MOSFET manufacturing technology usually leads to a high carrier lifetime. Therefore, a high stored charge and a high peak reverse current of the diode occurs in conventional MOSFETs. This is an impediment for many applications.

A carrier lifetime adjustment is applicable to reduce the stored charge. It has to be carried out in a separate production step. As a first approximation, the insertion of recombination centers in the n^--region does not affect the properties of the

MOSFET, because the MOSFET is a unipolar device. During the on-state of the MOSFET, carrier recombination cannot take place. Hence, the resistance R_{on} should remain unaffected. However, secondary effects have to be taken into account. Recombination centers, which are inserted for the reduction of the carrier lifetime, can retroact on the effective doping. Thus, the utilization of gold is ruled out, because gold, acting as acceptor, will compensate the base doping and increase the resistance R_{on}. This effect does not occur when platinum or electron irradiation is used. With electron irradiation it has to be considered that it affects the charge in the gate oxide. Electron irradiation reduces the threshold voltage V_T. By means of adequate annealing, the threshold voltage can partially be restored.

MOSFETs with platinum diffusion or electron irradiation, which are used for the reduction of the stored charge of the inverse diode, are known as "FREDFET" (Fast Recovery Diode Field Effect Transistor). Here, the stored charge of the inverse diode is reduced. The reverse recovery behavior is improved slightly, so that these diodes can be used in circuits with low parasitic inductance.

Primarily, the reverse recovery behavior is problematic. Though the area of the p-region is considerably smaller – a measure, which is also used with an MPS diode to achieve soft recovery behavior – fundamental requirements on the MOSFET are contradictory to what is necessary to achieve soft recovery turn-off behavior:

- To keep R_{on} as low as possible, the base of the MOSFET has to be designed as thin as possible.
- To obtain an effective short-circuit R_S, the p^+-doping is chosen as high as possible.

Both measures result in a snappy switching behavior of the diode, and they limit the possibility for both the MOSFET as well as the diode to be optimized.

In many hard-switching applications the inverse diodes are unusable, and they are often called parasitic diodes. By inserting a Schottky diode in series to the MOSFET and in reverse direction to the inverse diode, they can be suspended, and an optimized soft recovery diode can be connected in parallel. However, further losses occur due to the threshold voltage of the additional junction.

When a part of the current in the diode freewheeling mode flows via the n^+-source layer, the performance of the diode can be improved., as shown in [Zen00]. This effect is more pronounced in a Trench MOSFET [Dol04]. The effect is shown in Fig. 9.23. As already mentioned in the discussion of the threshold voltage at the beginning of the MOSFET chapter, a potential difference between gate and semiconductor exists due to the different positions of the Fermi level in the heavily n^+-doped poly silicon gate and in the p-type semiconductor, which acts as a small positive gate voltage. Further, if the current flow is in reverse direction, a potential drop builds up and the point D^* in Fig. 9.23 becomes negatively biased against the source. If $V_G = 0$ (=source potential), then this leads to a positive potential of V_G compared to D^*. This is of same effect as a positive voltage between gate and p-body, and an inversion channel can be formed, respectively the threshold voltage is dynamically reduced, as expressed by [Dol04]. Although the outside gate potential is set to zero, an electron-conducting channel becomes present. In devices

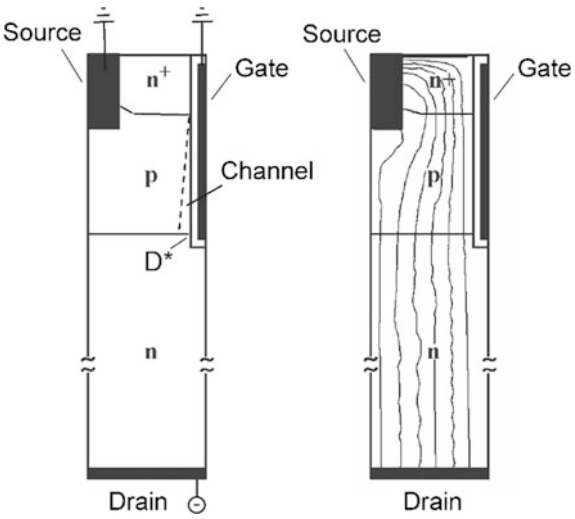


Fig. 9.23 Inverse diode in a trench MOSFET: potential and channel formation (left), simulated forward conduction current distribution (right)

with high channel density, as in Trench MOSFETs, it can become the dominant current component. Under these bias conditions, more than 90% of the total current is confined to the channel region [Dol04]. An example is given in Fig. 9.23.

The effect is somewhat similar to Fig. 9.8, where a positive potential at D* to source leaded to the channel pinch-off, now the potential at D^* is of opposite sign.

The occurrence of an electron current is of positive effect to the reverse diode characteristic. In its on-state, significant conduction takes place at a voltage below the conventional diode turn-on voltage at ca. 0.7 V, reducing conduction losses. Additionally, since this current is primarily due to majority carriers, this effect will positively impact the diode recovery behavior, since less stored charge will need to be removed during the reverse recovery process. Only a fraction of the total current is conducted by the pure diode part in the structure, i.e. the body-epilayer junction, as shown in Fig. 9.23.

Therefore, only the fraction of the total current which is conducted in parallel by the pure diode part causes an excess carrier concentration in the base-region of the pin-diode and contributes to the reverse-recovery charge. The ratio between the channel-conducted electron current part and the hole current injected by the body of the inverse diode depends on the device structure (thickness of gate-oxide, body doping). It also depends on the current density. The larger the current density, the larger becomes the share of injected holes and the larger will be the stored charge. With said measure, diodes of some modern Trench MOSFETs are significantly improved. Figure 9.24 shows a strong reduction of the stored charge of a new generation of trench MOSFETs (Device B) compared to an older design (Device A).

In applications with low blocking voltages and high currents (e.g. blocking voltage < 100 V and output current > 10 A), like switched power supplies, the technique of synchronous rectification is very common (see chapter MOS Controlled Diodes). pin-diodes or Schottky-barrier diodes used as free-wheeling

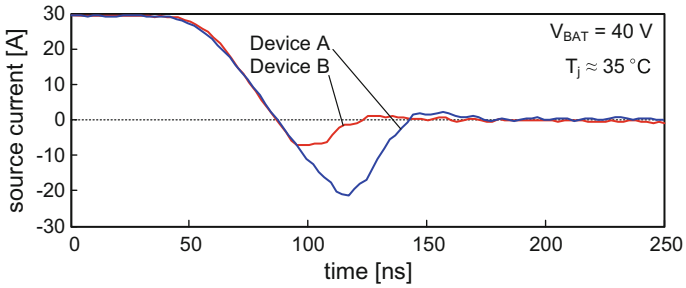


Fig. 9.24 Turn-on of MOSFETs in a half-bridge configuration with reverse recovery of the respectively inverse diodes. Both devices rated for 75 V and for $I_D > 100$ A. *Device A* IRF3808S planar technology, *Device B* IRFS3207 trench technology. *di/dt* = 800 A/µs

devices are replaced by MOSFETs, which are utilized for inverse conduction and allow a reduction of conducting losses due to the significantly lower voltage drop across the device.

When the channel of the synchronous rectifier is open, its intrinsic diode is bypassed. In the I-V-characteristic in Fig. 9.7 this mode is given by the dotted line in the 3rd quadrant for $V_G > V_T$. In this case, the voltage drop across the device is lower compared to the case of a closed channel, particularly when low currents are applied.

Figure 9.25 shows the circuit of a buck converter, which is a typical application of low-voltage power MOSFET. To avoid shoot-through currents, which would occur if the low-side MOSFET gate still has an on-signal while the high side MOSFET turns on, the gate control scheme for the switches must contain so-called interlock or dead times during the current commutation, wherein both channels are closed. In the example shown in Fig. 9.25, the low-side MOSFET must be turned-off before the high-side MOSFET is turned on. The same holds for the high-side switch. Thus, the intrinsic diode of the synchronous rectifier is conducting during these necessary dead times.

A control scheme for this type of operation is given in Fig. 9.26. During the diode conducting time, the gate of the low side MOSFET has the signal "on", and

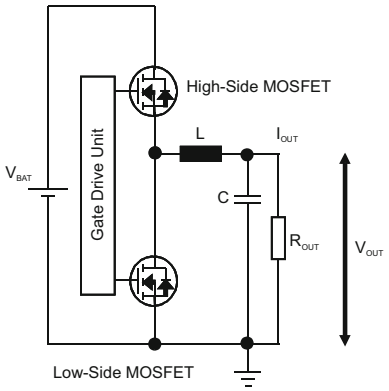


Fig. 9.25 Basic schematic of a buck converter with MOSFETs and their intrinsic diodes

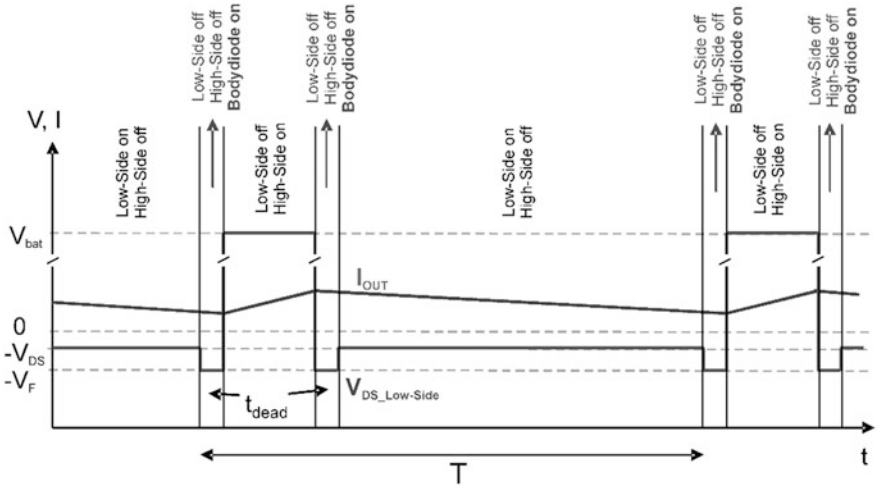


Fig. 9.26 Timing scheme for the buck converter as shown in Fig. 9.25

the voltage drop across the device is $-V_{DS}$, a value below the junction voltage of the intrinsic diode. During the dead times, the voltage drop is $-V_F$, a value which is in its absolute value above the junction voltage of the intrinsic diode.

Besides the effort of precise gate control, the turn-off behavior of the inverse diode becomes again one of the main drawbacks in this kind of application. In practice, the dead times cannot be reduced below a certain value, thus, the losses related to body diode conduction and body diode turn-off are responsible for an increasingly significant part of the total losses with increased switching frequencies.

The losses caused by the inverse diode of the MOSFET make up the largest share of the total losses in case of devices with low breakdown voltages. Inserting a Schottky-barrier diode in parallel to the synchronous rectifier is a way to reduce or to prevent the conduction of its body diode during the dead times. While the effect of this measure is limited by parasitic inductances of the circuit when discrete devices are used [Pol07], it shows significant improvement of the switching behavior if the Schottky barrier diode is integrated monolithically into the MOSFET die [She90, Cal04, Bel05].

9.12 SiC Field Effect Devices

9.12.1 SiC JFETs

SiC unipolar devices allow a very thin drift zone and a higher doping of the base, they attain a much lower resistance R_{epi} compared to Si, see Fig. 6.9. Therefore, research and development in SiC MOSFETs is done since several years. However,

mostly due to problems manufacturing a stable gate oxide, the first stable field effect device from SiC was the Junction Field Effect Transistor (JFET). The structure of a SiC JFET half cell [Mit99], which is in this configuration fabricated by Infineon, is shown in Fig. 9.25. With $V_G = 0$, there is a path from source to drain, and the structure is conducting. For the blocking mode, a negative voltage must be applied at the gate, this builds a space charge between gate and source, and the channel is pinched-off (Fig. 9.27).

The I-V characteristic of a JFET is shown in Fig. 9.28. For $V_G = 0$ V the JFET is in the forward conducting mode. With $V_G < 0$, the channel is narrowing, and for $V_G = -20$ V this sample is in the forward blocking mode. The cut-off voltage V_{CO} of this specific sample was at $V_G = -17.5$ V. To ensure reverse blocking of the JFET, a gate voltage below the cut-off voltage must be applied, however the negative gate voltage is limited by the breakdown voltage $V_{BD}(GS)$ of the gate-source diode, which occurs typically at −25 V. Due to tolerances in the manufacturing process, there is a spread of the values of V_{CO} as well as $V_{BD}(GS)$, this spread should be small. To ensure blocking capability, the JFET needs a negative voltage between V_{CO} and $V_{BD}(GS)$.

A "normally on" device is unwanted for voltage source converter applications. However, this can be solved by a MOSFET in series to the JFET in the cascode configuration, as shown in Fig. 9.29.

If a voltage $V_G > V_T$ is applied to the MOSFET, it is in the conduction mode. The normally-on JFET in series is conducting, and the cascode configuration is in the conduction mode. If the MOSFET is turned off, a voltage up to its blocking capability is built up. This voltage, which has a negative polarity to the source of the JFET, is applied to the JFET gate. If it is lower than the pinch-off voltage of the JFET, the JFET is turned off. The JFET is in the blocking mode. The behavior of the cascode configuration is similar as that of a MOSFET and it can be operated widely similarly as known for MOSFET.

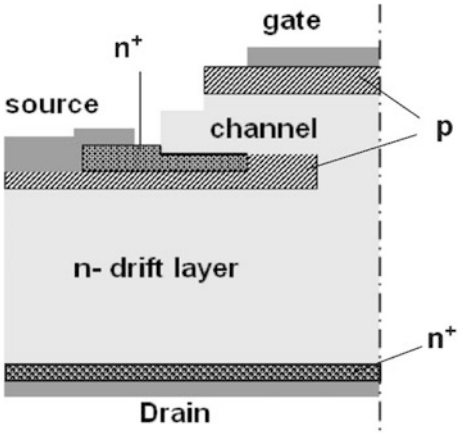


Fig. 9.27 SiC JFET half cell

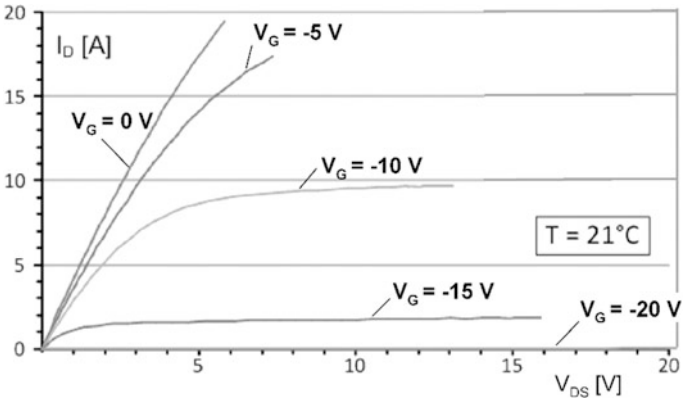


Fig. 9.28 Forward I-V characteristic of a JFET

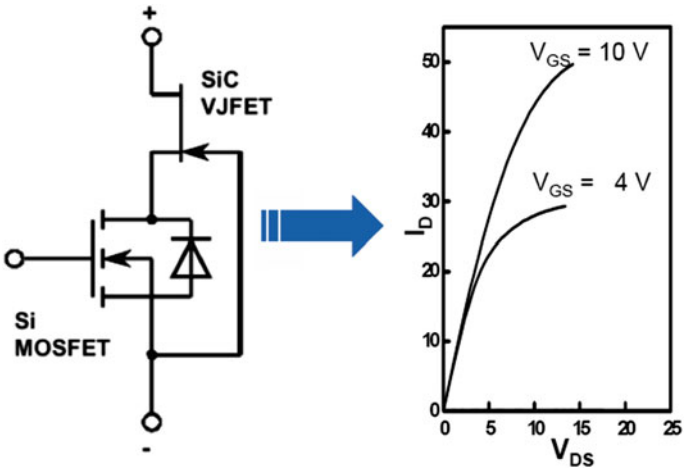


Fig. 9.29 JFET and low-voltage MOSFET in the cascade mode. Configuration (left), forward I-V characteristic of the cascode with V_{GS} (=V_G) at the Si-MOSFET gate (right)

In the conduction mode, the R_{on} of the MOSFET is in series to the R_{on} of the JFET. However, MOSFETs in the 30 V range can meanwhile be designed with extremely low on-resistances. They are available with R_{on} down to 0.1 mΩcm^2. Nevertheless, packaging an additional device requires additional effort.

Critical for the configuration in Fig. 9.29 is the control of the switching slopes at turn-on and turn-off. Especially, oscillations may occur during the cascode switching processes. They are caused by the capacitive elements of the cascode arrangements, which form resonant circuits with unavoidable inductive elements in the complete circuit. In particular, the turn-off process was found to be critical

[Sie12]. As an alternative, a so-called "direct-driven JFET" was proposed. Here separate drivers for the MOSFET and the JFET are present. The additional driver can be integrated into a single driver IC.

9.12.2 SiC MOSFETs

The SiC MOSFET is a very attractive device, and much progress in research and development has been achieved over the past decade. From several manufacturers, commercial released devices are available.

SiC MOSFETs are of n-channel type and have basically a similar structure as the vertical DMOS transistor (see Fig. 9.4). However, a problem of the SiC MOSFET is the channel conductivity, since the effective mobility of electrons in the channel is low. Whereas channel mobilities in the range of 5–10 cm^2/Vs were typical for first SiC MOSFETs, channel mobilities up to 13 $cm^2V^{-1}s^{-1}$ have also been reported [Ryu06]. It must be considered that also in Si MOSFETs the channel mobility is lower than the bulk mobility due to the influence of the surface (in the range of 500 $cm^2V^{-1}s^{-1}$, see Eq. 9.19), but the effect is much worse in SiC. This is due to a high density of electron traps in the SiC MOS interface, which results in the capture of channel electrons and in increased Coulomb scattering due to these captured electrons [Ima04]. The gate oxide is typically grown in an atmosphere containing NO or N_2O to reduce MOS interface state density.

According to (9.22), the total resistance R_{on} of the MOSFET has different contributions, where R_{ch} and the drift layer resistance R_{epi} are main contributors. For R_{ch}, Eq. (9.3) can be used with the parameters for SiC, and for R_{epi} Eq. (9.23) holds. For the dependency of R_{epi} on the blocking voltage, Eq. (6.16) derived for SiC Schottky diodes is suited.

R_{ch} was found to decrease with temperature for the first generations of SiC MOSFETs. The channel mobility was found to increase with temperature due to decreasing Coulomb scattering at surface states, however not to exceed 16 $cm^2V^{-1}s^{-1}$ at the highest temperature. While the channel mobility increases with temperature, the mobility in the drift region decreases keeping the on-state resistance almost independent of temperature [Rum09].

A measurement of R_{on} over time for two SiC MOSFETs available on the market is displayed in Fig. 9.30. It is seen that for more recent SiC MOSFETs R_{on} increases with temperature, however, from 25 to 125 °C only by 25%. For the Si-MOSFET, R_{on} usually doubles between 25 and 125 °C. This shows that there is a high contribution of R_{ch} to the total mobility.

In a more detailed treatment, the channel mobility μ_{ch} consists of different contributions and their influence can be combined with "Matthiessen's Rule" (developed from work by Augustus Matthiessen in 1864) which for the given situation can be written as [Uhn15]:

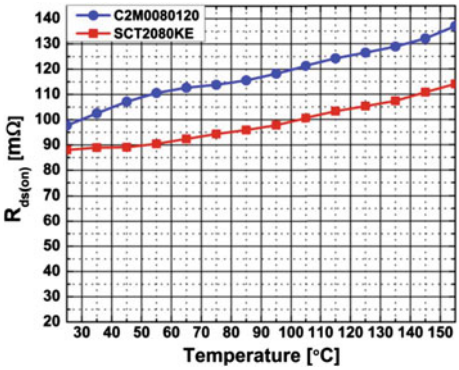


Fig. 9.30 On-state resistances versus temperature increase of the SiC MOSFETs SCT2080KE (manufactured by ROHM) and C2M0080120 (manufactured by Cree). Figure from [Muh16]

$$\frac{1}{\mu_{ch}} = \frac{1}{\mu_{bulk}} + \frac{1}{\mu_C} + \frac{1}{\mu_{sr}} + \frac{1}{\mu_{sp}} \tag{9.40}$$

Wherein μ_{bulk} is the bulk mobility, as expressed before as μ_n for the n-channel MOSFET, μ_C is the mobility contribution due to scattering at surface states, μ_{sr} due to surface roughness and μ_{sp} due to surface phonons.

The term with the lowest mobility is most strong in the result of Eq. (9.40). μ_C strongly increases with temperature, since the amount of charged traps decreases with temperature, and additional due to the increased Fermi-Potential more electrons are present shielding the scattering centers. Further, a strong dependency on the doping N_A at the position of the channel was found. For SiC MOSFETs with lower N_A, the contribution of μ_C is found to be smaller. For a lowly p-doping, channel mobilities up to 70 $cm^2V^{-1}s^{-1}$ are reported, and a decrease of μ_{Ch} with temperature was found [Uhn15].

In modern SiC-MOSFETs, further the channels are designed as very short, and additional effort is done to improve the channel conductivity e.g. by an additional thin n-doped layer.

Prototypes of SiC MOSFETs reached remarkable low R_{on}, e.g. 5 $m\Omega cm^2$ [Miu06] and 1.4 $m\Omega cm^2$ [Nak11] for a 1200-V type. However, these values are still far above the theoretically possible low limits for SiC (see Fig. 6.9). Intensive work is on-going. The structures of SiC trench-MOSFETs are shown in Fig. 9.28.

In a trench-MOSFET (Fig. 9.31 left), the resistance R_a is eliminated, however, at applied high drain-source voltage the highest electric field occurs at the gate oxide at the trench bottom. In the double-trench MOSFET structure (Fig. 9.31 right) the position of highest field moves to the edge of the source trench whose position is deeper. Hence, the gate trench is released from high electric fields to some extend [Nak11], preventing overload or destruction of the oxide layer at the gate trench.

Important for the reliability of SiC MOSFETs is the gate oxide quality. The commonly used material for gate oxide in Si and SiC devices is SiO_2. In some first

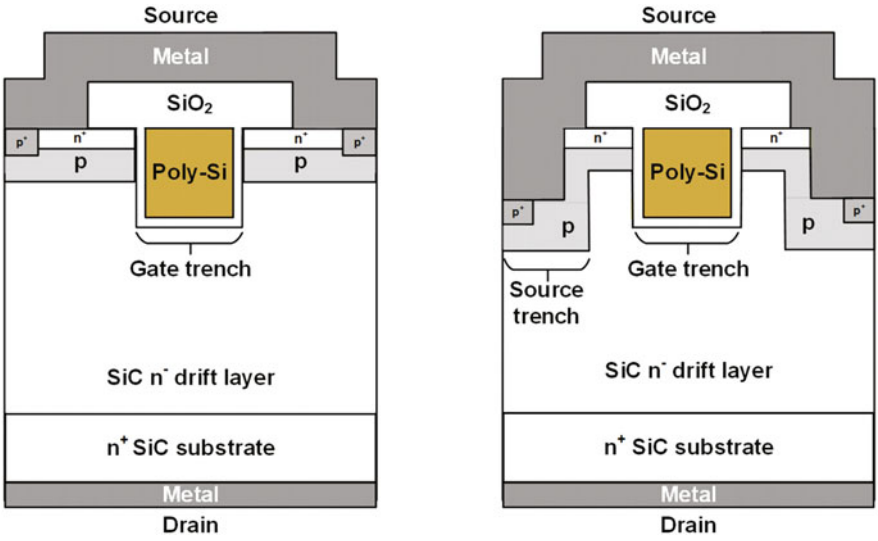


Fig. 9.31 Trench MOSFET and double-trench MOSFET structures. Figure according to [Nak11]

generations of SiC MOSFETs, the gate oxide is thinner in comparison to Si power MOSFETs and IGBTs due to the impact of oxide thickness on the channel resistance, see Eqs. (9.1) and (9.3). At a high electrical field, the Fowler-Nordheim tunneling which describes emission of electrons at surfaces in presence of high electric field [Fow28] contributes to dielectric breakdown [Scd96]. The Fowler-Nordheim tunneling current is higher in SiC than in Si [Gur08]. The tunneling current is exponentially dependent on the electric field in the dielectric and barrier height to carriers. This barrier height is primarily determined by band offsets between SiC and the dielectric. Since band offsets for SiC to SiO_2 are smaller than those with respect to Si, a lower reliability is expected for SiC MOS-based devices as compared to MOS devices for the same electric field oxide from Si [Sin04]. Further, in oxide layers on SiC exists a high density of interface and near-interface traps [Gur08].

It was shown that the effective lifetime of SiC MOSFETs will be determined not by oxide lifetime (so-called intrinsic failure) but by extrinsic failures which are caused by imperfections of the manufacturing process. Gate stress tests reported in [Bei16, Bei17] showed several extrinsic failures. Major differences between the investigated manufacturers were found; however, it is shown that SiC MOSFETs might reach IGBT-like gate oxide reliability and that they are on the way to become mature and reliable devices. It is possible to obtain SiC MOSFETs down to the same low ppm rate as Si MOSFETs or IGBTs by applying smart screening measures [Lut18]. The enabler for efficient gate oxide screening is a much thicker bulk oxide than what is typically needed to fulfill intrinsic life time targets.

A further challenge for the SiC MOSFET is the control of the threshold voltage V_T [Aga06]. In particular, V_T is found to shift with accumulated stress for the device. Possibly, a thick enough gate oxide and a suited screening process will also minimize this effect.

9.12.3 The SiC MOSFET Body Diode

Changing the polarity of V_{DS}, the inverse body diode of the SiC MOSFET is in the forward conduction state. However its threshold voltage is for devices of several manufacturers not at the expected threshold voltage of a pn-junction in SiC at about 2.7 V at room temperature, but at much lower values. Indeed, when $V_G = 0$ (=source potential) and a voltage drops across the body pn-junction, a positive potential of V_G appears at the p-body to n^--SiC interface. This effect is the same as if a positive potential of gate to p-body is applied such that and an inversion channel can be formed. This effect is already known from Si-MOSFETs, see Sect 9.10 and Fig. 9.2. However, due to the higher built-in voltage of a pn-junction in SiC, it is much more apparent in SiC. An example of the forward characteristics of the inverse diode for different gate voltages is shown in Fig. 9.32. A negative voltage at

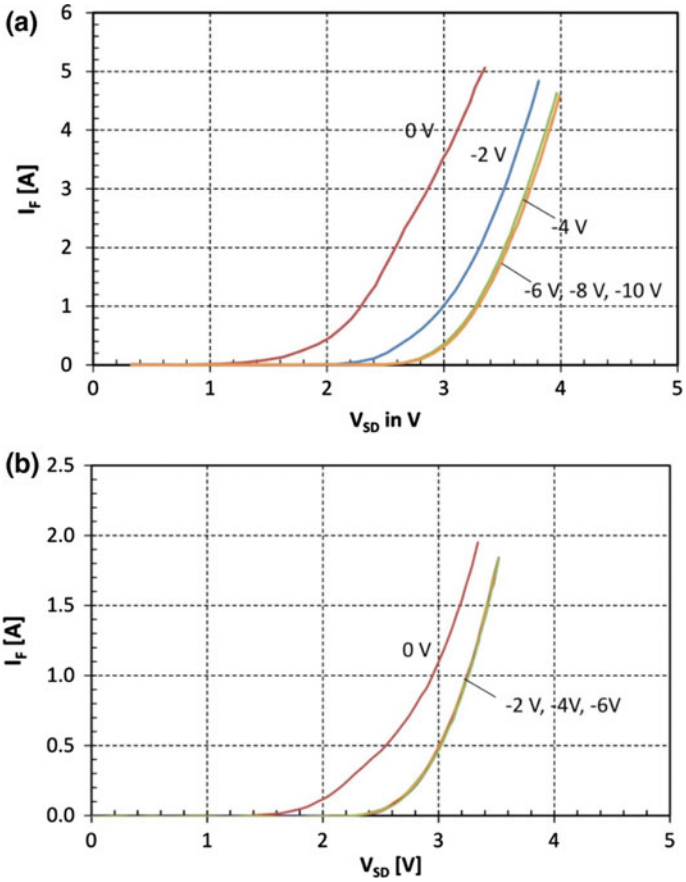


Fig. 9.32 Forward characteristics of the inverse diode of a SiC MOSFET for different gate voltages. **a**) 25 °C **b**) 150 °C

the gate of −6 V (RT) or −4 V (150 °C) must be applied before the channel is closed and the threshold voltage of the pn-junction becomes visible.

In first applications, a SiC Schottky diode was connected antiparallel to the MOSFET. More and more, the inverse MOSFET diode is used in the synchronous rectifier mode, compare chapter on diodes. In this mode, the inverse current is conducted by a significant part in unipolar mode. Short before turn-off, however, the channel must be closed to avoid phase-leg short circuit. Figure 9.32 shows that also in this mode the main part of the current is flowing across the channel, this is an advantage for the switching behavior, as explained in Sect. 9.11.

9.13 GaN Lateral Power Transistors

Vertical GaN Power devices are not available yet. All GaN devices used as power devices are lateral devices. Their function is based on the 2-dimensional electron gas, which connects source and drain and is switched by the gate.

For the lateral resistance with the length L and width W and a charge density per area G_{2DEG}, one can calculate (compare Eq. 9.3)

$$R_{on} = \frac{L}{W \cdot \mu_{2DEG} \cdot q \cdot G_{2DEG}} = \frac{L}{W \cdot \mu_{2DEG} \cdot Q_{2DEG}} \tag{9.41}$$

G_{2DEG} for AlGaN/GaN heterostructure can be above 10^{13} cm^{-2}, μ_{2DEG} can be up to 2000 cm^2/Vs. In a lateral device, the lateral electric field shape is rectangular in ideal case. Then holds for the breakdown voltage [Ued17]:

$$V_{BD} = \mathsf{E}_c \cdot L_{GD} \tag{9.42}$$

E_C is the critical field strength and L_{GD} the distance between gate and drain, see Fig. 9.33. For the length L in Eq. 9.41, the distance between gate and source L_{GS}

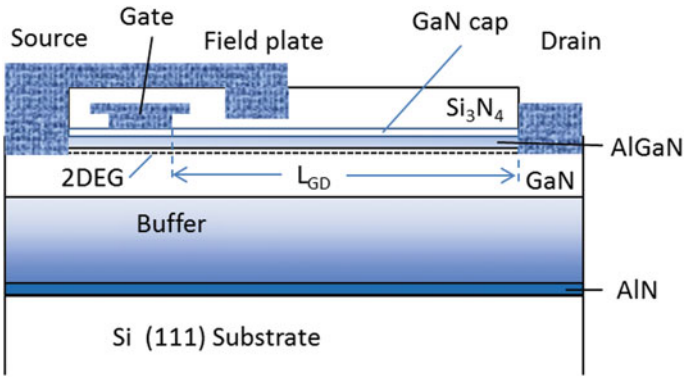


Fig. 9.33 GaN High-electron-mobility transistor (HEMT)

has to be added, $L = L_{GD} + L_{GS}$. Then from (9.41 with (9.42) it can be derived for R_{on} [Ued17]

$$R_{on} = \frac{L_{GS}}{W \cdot \mu_{2DEG} \cdot q \cdot G_{2DEG}} + \frac{V_{BD}}{W \cdot \mathsf{E}_c \cdot \mu_{2DEG} \cdot q \cdot G_{2DEG}} \tag{9.43}$$

For lateral devices the performance is often specified by resistance times gate width $R_{on} \cdot W$. For devices with blocking capability of 650 V, values as low as 27 Ωmm can be seen in recent publications [Miy15]. The resistance per area depends on details of the realized structure as well. High channel width W per area is desired. The structure is arranged in stripes, which shall be narrow. Due to the high critical electric field strength in GaN, the lateral extension L_{GD} is quite small, significant below 10 µm for a 650 V device. In contrast to lateral devices made in Silicon, GaN devices require a significantly smaller surface area for the space charge. The length L_{GS} is nowadays below 1 µm, e.g. 0.8 µm. Therefore, with GaN even in a lateral device a high current density in respect to the die area can be achieved. Comparing the resistance for the device area, today values of 0.54 Ωmm^2 [Miy15] and 0.66 Ωmm^2 [Hil15] have been demonstrated. These values are well beyond 1–2 Ωmm^2 which are achieved for nowadays with 650 V superjunction structures, as illustrated in Fig. 9.11.

Especially the gate capacities are much lower for GaN devices compared to Si. This leads to low switching losses and allows high switching frequencies. An often used criterion is the product $R_{on} \cdot Q_g$ with

$$Q_g = C_{iss} \cdot (V_{Gon} - V_{Goff}) = (C_{GS} + C_{GD}) \cdot (V_{Gon} - V_{Goff}) \tag{9.44}$$

Since the gate capacities are proportional to W, the product $R_{on} \cdot Q_g$ resp. $R_{on} \cdot C_{iss}$ becomes independent of W. With GaN, a $R_{on} \cdot Q_g$ is achieved about one decade smaller compared to the best Si MOSFETs in the 600 V voltage range.

Two main structures are developed today. The first, the high-electron-mobility transistor (HEMT), is initially a device for microwave application and was presented several decades ago with the GaAs/$Ga_{1-x}Al_xAs$ heterojunction [Mim80]. After the methods for deposition of GaN on sapphire substrates by MOCVD were developed, the production of AlGaN/GaN-based HEMTs was possible. The high polarization in the AlGaN/GaN heterostructure leads to a high charge density in the 2-dimensional electron gas (2DEG), see Sect. 4.11. Nowadays, HEMTs for power applications are fabricated on Si or SiC substrates. The basic structure of a HEMT as power device on a Si substrate is shown on example in Fig. 9.33.

It was shown by [Sai03] that 600 V blocking voltage could be achieved, and a current up to 850 A/cm^2 could be switched. In the blocking mode, the electric field drops laterally across the length L_{GD}, see Fig. 9.33. In [Sai03] L_{GD} amounts only 10 µm, in recent structures it can be even lower. The field plate on top in Fig. 9.33 expands the electric field from source in direction drain, the function is similar to a field plate in Fig. 4.25. Also multi-step field plates are in use [DeS16].

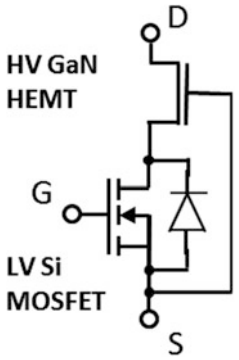


Fig. 9.34 GaN HEMT and Si MOSFET in cascode configuration

Meanwhile 600 V HEMTs are available on the market [Hon15]. To interrupt the conductivity of the 2-dimensional electron gas, a negative voltage at the gate must be applied in case of a depletion-mode HEMT. The device shows normally-on characteristics. To achieve normally-off characteristics, as preferred in typical power electronics voltage source converters, the HEMT device is combined with a low-voltage Si MOSFET in cascode configuration as shown in Fig. 9.34. The cascode is similar to the SiC JFET in cascode configuration shown in Fig. 9.27 and acts in the same way as explained in Sect. 9.12.1.

With this configuration, when a positive gate voltage is applied to the MOSFET, at the GaN HEMT the gate voltage drop equals the voltage drop between drain and source of the low-voltage Si MOSFET, which must be small to avoid losses. Hence, a relatively low positive gate voltage is applied at the gate of the HEMT.

For a long time, GaN devices suffered from the fact that they lacked current withstand capability at avalanche breakdown. In 2015, HEMT devices with avalanche capability have been reported [Liu15].

As a second main type, a normally-off 600 V Gate Injection Transistor (GIT) is available [Mor14, Ish15]. The basic structure is displayed in Fig. 9.35.

The GIT was first presented by [Uem07]. It features a p-AlGaN gate formed over the undoped AlGaN/GaN heterostructure, as shown in Fig. 9.35. The p-AlGaN lifts up the potential at the channel which enables normally-off operation. At the gate voltage of 0 V, the channel under the gate is fully depleted and no drain current is flowing. At the gate voltages between threshold voltage V_T and the forward built-in voltage V_{bi} of the pn junction, the 2D electron gas is recovered and the GIT is operated as a field effect transistor. Further increase of the gate voltage exceeding V_{bi} results in the hole injection from the p-AlGaN to the channel.

The injected holes accumulate the equal number of electrons that flow from the source to maintain charge neutrality at the channel. The accumulated electrons are moved by the drain bias with high mobility, while the injected holes stay around the gate, because the hole mobility is at least two orders of magnitude lower than that of the electrons. This kind of local conductivity modulation results in an increase of the drain current [Uem07]. In the $g_{fs} - V_G$ characteristics this occurs as a second peak, as shown in Fig. 9.36. However, since the holes do not contribute to current

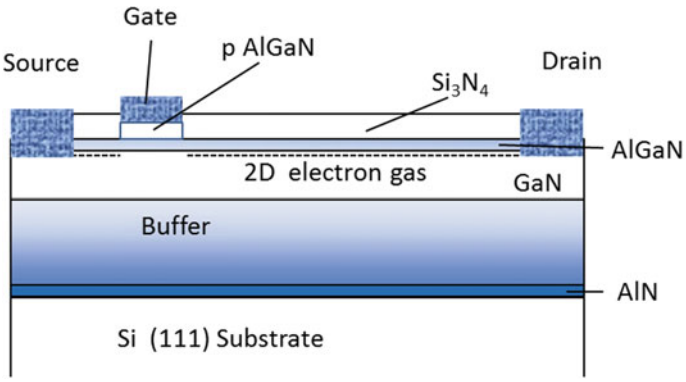


Fig. 9.35 Gate injection transistor (GIT) at $V_G = 0$ V

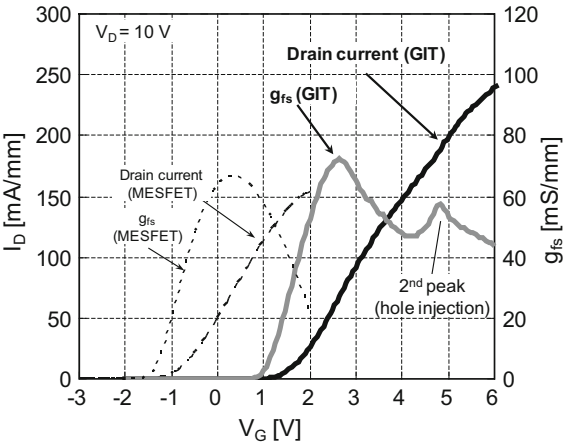


Fig. 9.36 $I_D - V_G$ and $g_m - V_G$ characteristics of **a**) a fabricated GIT and **b**) a MESFET (metal semiconductor field effect transistor). The GIT shows a peculiar transconductance characteristics with two peaks. Figure according to [Uem07]

transport, the GIT is not treated as bipolar device. The positive gate is usually limited to 5 V in order to keep the injected gate current below 10 μA/mm [Hil15].

The gate threshold voltage of a GIT is typically in the range of only 1–1.2 V. In applications with high electromagnetic noise, this low V_T could be a problem concerning unwanted turn-on, leading to short circuits in converter phase-legs. By introducing a negative gate voltage at the gate driver, GITs can be operated in converter bridges without shoot-through currents [Mor14].

Further concepts for normally-off (enhancement mode) GaN HEMTs are displayed in Fig. 9.37. Figure 9.37a shows the "recessed gate" structure, in which the thickness of the AlGaN layer is reduced. The threshold voltage becomes positive, is however typically below 1 V. In Fig. 9.37b an additional insulator layer is implemented, leading to a reduced gate leakage current. Different insulators are reported, oxides such as Al_2O_3 or HfO_2 and are possible [Hil15]. Figure 9.37c displays the GaN Metal Insulator Semiconductor Field Effect Transistor (MISFET).

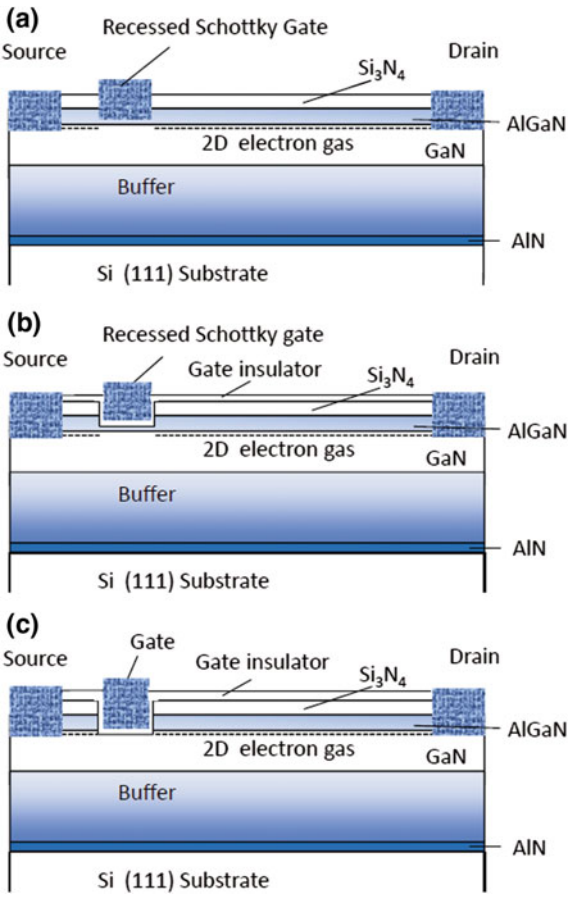


Fig. 9.37 Structures for enhancement mode GaN devices. **a** Recessed Schottky gate **b** recessed Schottky gate with additional gate insulator **c** GaN MISEFT. Figure inspired by [Wür15]

There is a full gate recess through AlGaN barrier. The insulator can consist of SiO_2 [Wür15] or a second thin layer of Silicon Nitride, in [Hua17] an additional interface protection layer of oxidized GaN is applied. At a positive voltage at the gate, an electron accumulation is created in the low conductive GaN layer below the gate, leading to a lateral conductivity.

A further method to achieve enhancement mode is the implementation of fluorine into the AlGaN layer below the gate, see Fig. 9.38. Since the group-VII element fluorine (symbol F) has the strongest electronegativity among all the chemical elements, a single F atom at the interstitial site tends to capture a free electron and becomes a negative fixed charge. These negative fixed charges deplete the two-dimensional electron gas (2DEG) in the channel at $V_G = 0$ V [Che17]. To restore the 2DEG a positive gate voltage is required. A threshold voltage up to +3 V is possible [Wür15]. The F^- incorporation can be combined with measures shown in Fig. 9.37, for example a second SiN_x thin film is applied as gate insulator

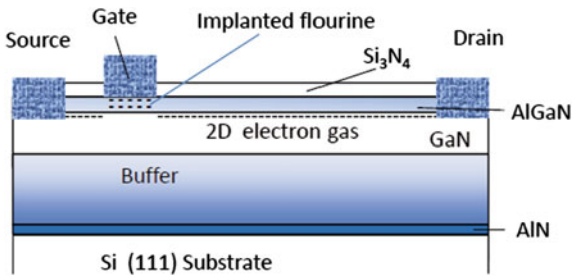


Fig. 9.38 GAN HEMT with negatively charged fluorine ions below the gate

to reduce the gate leakage current. With this combination, a threshold voltage of +3.6 V is achieved in [Che17].

The normally-off type is favored for power electronic applications. There are different approaches, a variety of different concepts, and no clear commitment to a specific technology. Measures shown in Figs. 9.35, 9.37 and 9.38 can also be combined. High threshold voltage levels (> +3 V) combined with low gate leakage current is desirable [Wür15]. Intensive work in research and development is done.

A problem of all lateral GaN devices is the so-called **current collapse**, this is the temporarily decrease of the 2DEG conductivity after applying blocking voltage. It leads to a decreased current capability I_{Dsat}. Power devices are usually not operated in the linear mode, however the effect of decreased 2DEG conductivity results also in the increase of the on-state resistance R_{on}. If desaturation occurs, the device might no longer be capable to carry the applied current, therefore the denomination as current collapse. More recently, the denomination "**dynamic R_{on}**" is often used for this phenomenon. The effect is shown in Fig. 9.39.

The dynamic R_{on} effect is reversible. It depends strongly on the value of the blocking voltage which is applied before turn-on, and on the time interval of negative voltage. For high switching frequency, the blocking time intervals are short and the effect is less exposed.

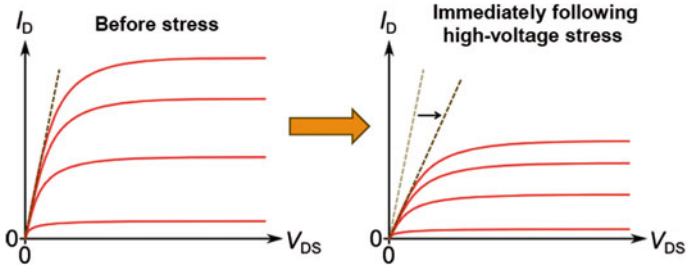


Fig. 9.39 Exemplifying drawing of the current collapse respectively dynamic R_{on} for a GaN device. Figure from S. Sque, NXP Semiconductors [Squ13]

Regarding its explanation, one has to consider that the Si-Substrate (Figs. 9.33, 9.35, 9.37, 9.38) is usually at the same potential as the source electrode. So the applied voltage lies in the same way across the vertical layers. The vertical blocking capability is specified in GaN-on-Si epitaxial wafers to 1000 V [EPI16]. A part of the leakage current will flow between substrate and drain.

There are two explanations:

- At off-state blocking, electrons from the gate are injected into trap states next to the gate. At on-state after stress, trapped electrons act like a negatively biased gate. The 2DEG is partially depleted leading to increased R_{on}. Time-dependent negative charges de-trap and the 2DEG current capability is restored.
- In blocking state, the Si-Substrate is usually at the same potential as the source electrode. Thus, the applied voltage also stands across the vertical layers. A part of the leakage current will flow between substrate and drain. Electrons are trapped in the bulk forming negative charge states. Trapped electrons partially deplete the 2DEG above. After the electrons de-trap, the 2DEG current capability is restored.

Lots of work is done for improvement. It seems that the main effect is the creation of negative charges by the leakage current from the substrate: An effective measure for the GIT is reported as the so-called Hybrid-Drain-embedded GIT, the HD-GIT. Its features an additionally p-GaN region formed in the vicinity of the drain, see Fig. 9.40 [Kan15]. This p-GaN region is electrically connected to the drain electrode by the interconnection metal layers. Injected holes from the p-GaN at the off-state effectively release the trapped electrons so that the current collapse is fully eliminated. In order to avoid the depletion of the 2DEG under the p-GaN region, a thicker i-AlGaN layer is employed, while at the gate a recessed gate structure is applied, compare Fig. 9.37a.

GaN device technology is still at a very early stage of development. The devices compete with Si superjunction devices in the range of 650 V and above. It is

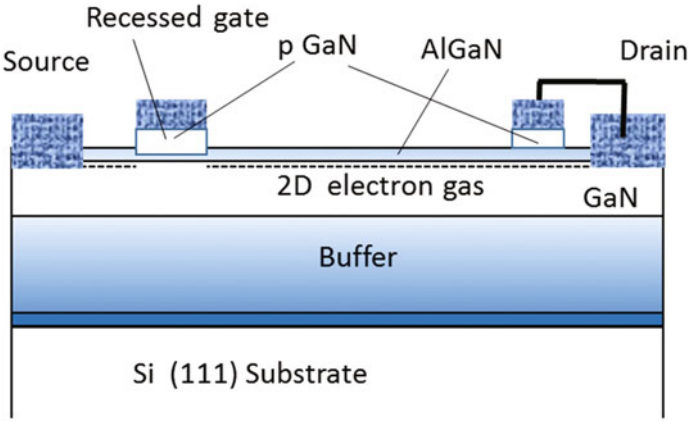


Fig. 9.40 Hybrid-Drain-embedded GIT (HD-GIT). A p-GaN at the drain and a recessed-gate structure are introduced in the HD-GIT. Figure following [Kan15]

expected that the main application field for GaN is the voltage region below 1000 V. The parasitic capacities C_{GS} and C_{GD}, summarized as C_{iss}, are much smaller compared to the superjunction devices [Hil15], enabling higher switching slopes and higher switching frequencies (above 1 MHz). Therefore these devices can lead to very compact and efficient power electronic converters. However, the reliability for industrial applications and electromobility applications is to be proven.

9.14 GaN Vertical Power Transistors

As a promising vertical device from GaN is the current aperture vertical electron transistor (CAVET) is described [Cho13]. Its structure is shown in Fig. 9.41.

The horizontal high-mobility electron channel at the interface of the AlGaN/GaN is located beyond the Gate electrode. A vertical GaN drift region is arranged through the aperture with the width L_{ap}. The buried current blocking layers, denoted as CBL in Fig. 9.41, consist of p-doped GaN. Since the structure sustains the voltage in vertical direction, a high blocking voltage can be achieved with a small chip area.

The development of vertical GaN devices, however, is in a very early stage. Realized devices reached 200 – 300 V, with high leakage current. The devices are of normally-on type and require a need negative gate voltage for blocking [Cho13]. Nevertheless, vertical GaN devices are promising as power devices due to the material characteristics of GaN.

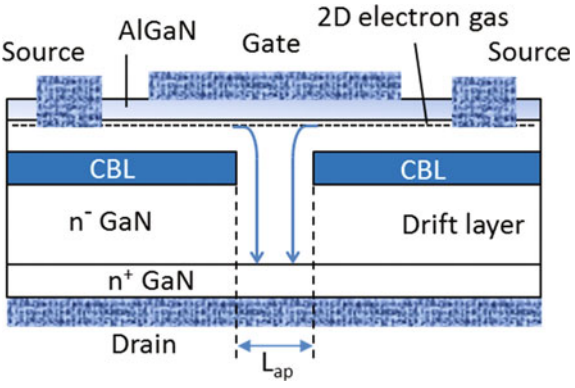


Fig. 9.41 Hybrid-current aperture vertical electron transistor (CAVET) fabricated with GaN. Figure following [Cho13]

9.15 Outlook

The MOSFET is a unipolar device. In forward direction, a "knee", i.e. threshold forward voltage does not occur. The MOSFET offers many advantages: its control is easy and requires only low power. The switching slopes are adjustable via gate resistances. At turn-off no tail current exists, in difference to IGBTs. The MOSFET has minor switching losses and high switching frequencies are possible. Moreover, it is short-circuit resistant and possesses a rectangular safe operating area.

Therefore, a MOSFET will always be the preferred device in applications whenever it is possible to apply it. MOSFETs can be paralleled without problems and the series connection is possible as well. Even high-power applications (50 kV, some kA), which require fast switching behavior and where costs are of minor importance, have been realized with parallel and series connection of MOSFETs.

A disadvantage of the Si-MOSFET, however, is the inverse body diode, which has insufficient properties.

When designed for higher blocking voltages, the resistance R_{on} increases strongly. The introduction of the superjunction principle, which does not show this strong correlation, represents an important development. Nowadays, superjunction MOSFETs are available for 600 and 900 V, even 1000 V devices are possible in principle. But the technological complexity is increases at higher voltage levels. Consequently, applications requiring higher voltages are still dominated by bipolar devices.

With superjunction devices in the range of 600 V, it is expected that future generations will achieve a further reduced R_{on} per area.

In the low voltage range (< 100 V), the trench technology leads to a reduction of the on-state resistance R_{on}. Further development in microelectronics will allow to produce finer structures and, thus, a higher cell density will be possible. This will further reduce the on-state losses. Furthermore, measures for the reduction of the capacitances are taken to reduce switching losses.

Field controlled devices made of SiC become available. Due to the possible very thin drift zone and the feasible higher doping of the base, they attain a much lower resistance R_{epi}, see Fig. 6.8. They extend the range of unipolar field controlled devices well above the range of 1000 V, several kV are possible. Also below 1000 V they will compete with Silicon devices, because of their potential to offer low R_{on}. SiC MOSFETs for 3.3 kV are announced, SiC MOSFETS of 10 kV and more are in development.

In the voltage range below 1000 V, lateral GaN transistors compete with Si MOSFETs. With GaN, low switching losses are possible, and a $R_{on} \cdot Q_g$ is achieved that is one decade lower compared to Si MOSFETs. Therefore GaN devices are very attractive for high-frequency applications. Many effort is done today to investigate the reliability for industrial applications and electromobility applications.

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